

Spectrum-Matched Quantum-Assisted PINNs: A Constructive Design Methodology for Problem-Specific Variational Circuits, Instrumented by Explainable-AI Diagnostics

Katta Lahari

Mannava Daasaradhi

September 2026

Abstract

Physics-informed neural networks (PINNs) learn low frequencies fast and high frequencies slowly (spectral bias), because their neural tangent kernel (NTK) spectrum decays with frequency. A data-re-uploading quantum circuit behaves differently: its output is a Fourier series whose frequency set is fixed, exactly, by its encoding. We join the two facts into a constructive design rule, Spectrum-Matched Circuit Design (SMCD): read the frequencies a PDE’s solution needs from the operator’s Fourier symbol and the forcing, then build the circuit whose frequency set covers them, with no architecture search. The hybrid model mixes K copies of the designed circuit through a linear head. Six explainable-AI instruments, run identically on matched classical and hybrid models, test the mechanism rather than only the accuracy, and every claim is pre-registered with a numeric threshold and scored by a script. The circuit produces exactly its designed frequencies (leakage below 4×10^{-15}). Designed frequencies beat random ones of the same size on the heat equation ($6.5\times$, one copy), on Poisson ($236\times$, eight copies) and on a groundwater problem ($9.9\times$, eight copies), on every seed. On waterlogging of farmland from leaking canals, the eight-copy circuit reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and places the waterlogged stretch to within 1 m. The instruments explain why copies matter: in one circuit every Fourier amplitude depends on the same angles, so the optimiser trades the slow wave for the fast one, and a single copy loses to classical baselines on Poisson. Two results bound the claim: on Poisson a classical model given the same frequencies is $3.3\times$ more accurate than the eight-copy circuit, and at high Helmholtz wavenumbers no model, classical or quantum, learns the solution.

1 Introduction

A classical physics-informed neural network (PINN) learns low-frequency solution components fast and high-frequency components slowly — spectral bias — because its neural tangent kernel (NTK) eigenspectrum decays with Fourier-mode index. Separately, a data-re-uploading variational quantum circuit realises a truncated Fourier series whose frequency set is determined exactly by the eigenvalues of its data-encoding Hamiltonians [10]. Prior hybrid quantum PINN work [2, 6, 11, 14] reports accuracy on hand-picked circuit sizes, without a principled way to choose the circuit *given* the PDE.

This paper joins the two facts. Given a PDE, the target spectral support is computed from the operator’s Fourier symbol and the forcing and boundary data; the circuit’s encoding scalings, re-uploading depth, qubit count and entangler then follow by formula so that its frequency set covers that support. We call this Spectrum-Matched Circuit Design (SMCD, Section 3). The hybrid

model mixes K copies of the designed circuit through a linear head: every copy has the designed frequencies, and the head sets their amplitudes independently.

A design rule is only useful if it can be tested, so the project is built around measurement. Six explainable-AI instruments run at the same checkpoints on matched classical and hybrid models (Section 4), so any difference can be traced to the quantum layer. Every claim about the method was written down with a numeric threshold and committed to git before the runs that test it, and a script issues every verdict.

Contributions.

1. **SMCD, a constructive map from PDE to circuit** (Section 3): four propositions and an algorithm that output the encoding scalings, depth, qubit count and entangler deterministically, with a design card that can be checked before training.
2. **A hybrid NTK decomposition** for models with disjoint classical and quantum parameters, $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q$, with no cross term (Section 3.5).
3. **Circuit copies**, a model family in which K copies of the designed circuit share its frequency set but have independent amplitudes (Section 3.8).
4. **An explainable-AI protocol for hybrid PINNs**: six instruments with published metric definitions, run identically on every model family (Section 4).
5. **Evidence** from seven model families on six PDE instances and one applied problem, groundwater under irrigated farmland (Sections 5–7).

Findings in brief. The circuit realises exactly the frequencies SMCD designs. Designed frequencies beat random ones of the same size wherever the circuit learns the solution: heat, Poisson and groundwater. One copy is not enough on problems that mix a slow and a fast wave, and the instruments show why; eight copies fix it, reaching 0.09% error on the groundwater problem, 18× better than a classical network of the same size. The method does not beat every classical model: on Poisson a classical model handed the same frequencies stays ahead, on the smoothing heat equation no model beats a plain MLP (as SMCD predicts), and at high Helmholtz wavenumbers every model fails.

2 Background

Full derivations for every item below are in `docs/derivations/`; this section states results and the one-line reason each matters to the design rule in Section 3.

2.1 Classical side

PINNs as strong-form collocation. A PINN [9] minimizes the PDE residual directly at sampled collocation points via automatic differentiation, with no mesh and no weak form — unlike Galerkin/FEM, the optimization problem is non-convex in the network parameters even when the underlying PDE is linear.

Exact derivatives via reverse-mode AD. Reverse-mode AD gives derivatives that are exact (to floating-point precision), not finite-difference approximations; the dominant cost for a PINN is second derivatives (e.g. u_{xx} in ≥ 2 dimensions), computed via double backward passes.

The NTK and lazy training [5] (Section 3.5’s classical baseline). Writing $\Theta(x, x') = \nabla_{\theta} N[u_{\theta}](x) \cdot \nabla_{\theta} N[u_{\theta}](x')^{\top}$ for the residual network’s tangent kernel, gradient-flow training linearizes in the lazy regime to $\dot{e} = -\Theta_0 e$, so each eigenmode of Θ_0 decays as $\exp(-\lambda_i t)$: loss-term imbalance between residual and boundary blocks is an eigenvalue-imbalance phenomenon, not merely a loss-weight-tuning problem (docs/derivations/03_ntk_pinn.md).

Spectral bias. For a tanh or ReLU MLP, the NTK eigenvalues decay roughly as a power law in Fourier-mode index, so high-frequency solution components are learned last and slowly (Figure 1, measured on the Poisson problem studied here). Fourier-feature and random-feature encodings [12] are the standard classical fix and a necessary baseline (Section 4.2): if the quantum layer beats only a plain MLP and not a Fourier-feature network, it has bought nothing that Fourier features do not already supply.

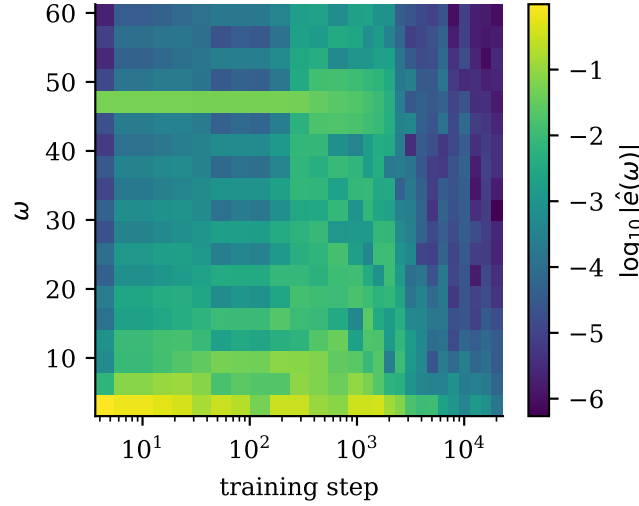


Figure 1: Spectral bias, measured: per-frequency error of a plain MLP over training on the Poisson problem, $u^* = \sin(\pi x) + \alpha \sin(15\pi x)$ with $\alpha = 0.05$. The $\omega = \pi$ mode converges first; $\omega = 15\pi \approx 47$ needs 16.2 times as many steps to reach tolerance (3 seeds).

Hard vs. soft boundary enforcement. The ansatz $u = B(x) + D(x) \cdot N(x)$ (distance function D vanishing on the boundary, particular solution B matching the boundary data) removes the boundary-loss NTK block entirely by construction, at the cost of a distance-function derivation per domain; the soft-BC alternative keeps a separately-weighted boundary loss term whose own NTK block this project instruments directly (Section 4).

2.2 Quantum side

Data-re-uploading circuits as Fourier series. For encoding gates $\exp(-ixH^{(l)})$ with generator eigenvalues $\{\lambda_a^{(l)}\}$, an L -layer re-uploading circuit’s output expectation is a truncated Fourier series whose frequency set is the set of signed sums of per-layer eigenvalue differences — the central theorem this project’s entire methodology is engineering on top of (Proposition 1, Section 3.2; 10).

Parameter-shift derivatives. $\partial_{\theta} f = \frac{1}{2}[f(\theta + \pi/2) - f(\theta - \pi/2)]$ for a Pauli-generated gate; because the circuit’s input x enters as a rotation angle, $\partial_x f$ is *also* exactly computable via a parameter shift — a hardware-exact PINN residual gradient, not a finite-difference approximation, and the reason this project’s whole approach is physically realizable on real quantum hardware,

not just simulable.

Barren plateaus. $\text{Var}[\partial_\theta C]$ decays exponentially in qubit count n for a global-observable cost function on a deep, unstructured random circuit [7]. SMCD’s Algorithm 1 (Section 3.6) mitigates this structurally: shallow depth (by design, not by luck — the depth rule targets the *minimum* L that covers the target band), a local observable, and small-angle initialization.

Effective dimension. The Abbas et al. effective-dimension statistic [1], computed from the empirical Fisher information eigenspectrum, is this project’s quantum-native capacity measure (Section 4’s Fisher/effective-dimension instrument), reported alongside — not instead of — parameter count, since “fewer parameters” is a claim prior QPINN work makes loosely.

3 Methodology: Spectrum-Matched Circuit Design (SMCD)

3.1 Setup

The hybrid ansatz used throughout (the *serial* form; Section 5 ablates parallel and random-scaling variants) is

$$x \xrightarrow{E_\varphi} z \in \mathbb{R}^m \xrightarrow{Q_\theta} \phi(z) \in \mathbb{R} \xrightarrow{w} u(x),$$

where E_φ is an affine classical encoder, Q_θ a data-re-uploading variational quantum circuit, and w a linear head. The split is deliberate and is itself a claim: the classical part learns the coordinate *warping*, the quantum part supplies the Fourier *basis*, and the head learns the *coefficients* — exactly the structure of a spectral method with a learned coordinate map.

3.2 Proposition 1 — Circuit spectrum

Proposition 1 (Circuit spectrum). *An L -layer data-re-uploading circuit, with encoding gate $\exp(-ix H^{(l)})$ at layer l for Hermitian generator $H^{(l)} = \sum_a \lambda_a^{(l)} |a\rangle\langle a|$, realizes an observable expectation of the form*

$$f(x) = \sum_{\omega \in \Omega} c_\omega(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L \left(\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)} \right) \right\},$$

i.e. a trigonometric polynomial whose frequency support is the set of all signed differences of encoding-generator eigenvalues, summed over layers.

Proof sketch. Expand $f(x) = \langle 0|U^\dagger(x, \theta) M U(x, \theta)|0\rangle$ by inserting the eigenbasis resolution of each diagonal encoding gate independently on the bra and ket sides. Every (bra-path, ket-path) pair contributes an x -dependent phase $\exp(ix \sum_l (\lambda_{a_l}^{(l)} - \lambda_{b_l}^{(l)}))$ multiplied by an x -independent amplitude collecting the trainable gates and M along that path; collecting paths by total frequency gives the boxed form. The full derivation, worked first for one qubit/one layer and then generalized to n qubits and d -dimensional inputs, is in docs/derivations/07_circuit_fourier_spectrum.md. ■

Two subtleties that change the constructive guarantee. Two facts, easy to get wrong and verified numerically against both an independent statevector simulator and a PennyLane oracle (agreement $< 10^{-10}$) before being trusted:

1. **Generator-vs-frequency factor of two.** The generator’s eigenvalues are $\pm\omega/2$ for an $RZ(\omega x)$ gate, but the *realized* frequency in $f(x)$ is the full ω , not $\omega/2$: the eigenvalue-difference structure above supplies a compensating factor of two. Reading the generator’s

eigenvalues off the gate and reporting that as the frequency halves every number in the design.

2. **State-preparation degeneracy.** If the qubit is prepared exactly at the equator ($RY(\pi/2)$ before the first encoding gate, an equal superposition), every frequency whose balanced-ternary leading digit is zero — exactly one third of Ω — has identically zero amplitude for *every* trainable-parameter setting, independent of ansatz expressivity. This is not a numerical inconvenience: it makes Proposition 3 below false for any target spectrum touching a dead frequency under the literal $\pi/2$ prescription. We use $\phi_{\text{prep}} = \pi/3$ throughout, verified to have no dead frequencies.

3.2.1 Corollary: ternary scaling and the depth rule

Set the per-layer scalings to $\omega_l = \Delta \cdot 3^{l-1}$ (single-wire case). Then $\Omega = \Delta \cdot \{-\frac{3^L-1}{2}, \dots, \frac{3^L-1}{2}\}$ — exactly 3^L distinct frequencies with no degeneracy, a bijection with the balanced-ternary representation of the integers in that range (proof: substitute $m_l = d_l - 1$ to convert the signed sum into an ordinary base-3 expansion shifted by $\frac{3^L-1}{2}$). This grows *exponentially* in depth, versus $2L + 1$ distinct frequencies for linear scaling $\omega_l = \omega_0$ — the design knob almost every prior re-uploading paper leaves fixed.

To guarantee the reachable band covers a target maximum frequency K at ternary spacing Δ :

$$\max |\Omega| \geq K \iff \Delta \frac{3^L-1}{2} \geq K \iff L \geq \log_3 \left(\frac{2K}{\Delta} + 1 \right) \implies \boxed{L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil}.$$

Both the factor of two inside the logarithm and the $+1$ matter. For the Poisson target spectrum ($K = 15\pi$, $\Delta = \pi$), the naive $L = \lceil \log_3(K/\Delta) \rceil$ gives $L = 3$, whose reachable band tops out at $13\pi < 15\pi$ and silently misses the target mode; the rule above gives $L = 4$, whose band reaches $40\pi \geq 15\pi$.

3.3 Proposition 2 — PDE target spectrum

Proposition 2 (PDE target spectrum). *For a linear constant-coefficient operator $\mathcal{L}$ with Fourier symbol $\sigma(k)$, the solution of $\mathcal{L}u = f$ on a periodic/rectangular domain satisfies $\hat{u}(k) = \hat{f}(k)/\sigma(k)$. The essential spectral support $S_\varepsilon = \{k : |\hat{u}(k)| > \varepsilon \|\hat{u}\|\}$ is therefore computable before training, from $\hat{f}$ and σ alone.*

For Helmholtz, $\sigma(k) = k^2 - |\mathbf{k}|^2$, so S_ε concentrates near $|\mathbf{k}| \approx k$ — the near-resonant band a classical PINN’s low-frequency-first learning dynamics reach last, if at all. For the heat equation, σ grows quadratically with $|\mathbf{k}|$ and the solution decays exponentially in time at every nonzero mode, so S_ε collapses toward $k = 0$ as t increases: this is the project’s negative control (Section 5.2) — there is no high-frequency content for a quantum layer to supply, and SMCD predicts zero benefit *a priori*, not just observes it after the fact.

3.4 Proposition 3 — Matching condition and resource bound

Proposition 3 (Matching & resource bound). *If $S_\varepsilon \subseteq \Omega$, the hybrid model represents the solution to ε accuracy with a linear head — the remaining optimization problem is convex in w . Covering S_ε with maximum frequency K and resolution Δ requires $L \geq \log_3(2K/\Delta + 1)$ layers (ternary scaling, above) and $n \geq \lceil d \rceil$ qubits for d input dimensions, plus ancilla wires for any required cross-frequency terms.*

This turns architecture choice into arithmetic: given (K, Δ, d) read off the PDE’s own symbol and data, (L, n) follow by formula, not by search.

3.5 Proposition 4 — NTK consequence

Proposition 4 (Hybrid NTK decomposition). *For a hybrid model whose parameters $\theta = (\theta_{\text{cl}}, \theta_q)$ partition into disjoint classical and quantum groups,*

$$\Theta_{\text{hyb}}(x, x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x'),$$

*exactly, with **no cross term**; a term of the form $2\Theta_{\times}$ does not arise for a shared-output, disjoint-parameter model.*

Proof. $\Theta(x, x') = \sum_{k=1}^P \partial_{\theta_k} f(x) \partial_{\theta_k} f(x')$ is a single inner product of gradient *vectors*, summed over the *full* index set of parameters $k = 1, \dots, P$. When that index set is partitioned into two disjoint blocks θ_{cl} and θ_q (a hybrid model’s classical and quantum parameters never overlap), the sum splits additively over the partition:

$$\Theta_{\text{hyb}}(x, x') = \sum_{k \in \text{cl}} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') + \sum_{k \in q} \partial_{\theta_k} f(x) \partial_{\theta_k} f(x') = \Theta_{\text{cl}}(x, x') + \Theta_q(x, x').$$

There is no third term: a cross term would only arise from a fundamentally different quantity, such as a covariance between two *separately* defined kernels or the square of a sum of outputs — not from the NTK of a single model with disjoint parameter groups, which this project’s ansatz is. The identity is also checked numerically on random synthetic Jacobians of matched and mismatched sizes; `src/qapinn/xai/ntk.py` implements it. ■

Because Ω (the quantum block’s realized frequency set) is discrete and prescribed by construction, Θ_q has eigen-directions *at* the encoded frequencies with eigenvalues bounded below independently of $|n|$ for $n \in \Omega$ — the exponential/power-law eigenvalue decay in $|k|$ that drives classical spectral bias (Section 2) is replaced by a flat response over Ω . This is the formal statement of “the quantum layer removes spectral bias inside its band,” and it is what Section 5’s NTK spectrum figures measure directly: decay exponent inside Ω versus outside it, for matched classical and hybrid models.

3.6 Algorithm 1 — SMCD

The design card (final step) is what makes the methodology auditable: every run’s predicted coverage can be set against its measured error (Section 5.4).

3.7 Octave-split circuit ensembles

When a single circuit cannot cover $\hat{S}$ within the barren-plateau depth budget ($L > L_{\text{max}}$), SMCD falls back to M shallow circuits, each matched to one octave band $[2^j \Delta, 2^{j+1} \Delta)$ of $\hat{S}$, summed at the head. This keeps every individual circuit shallow (trainable) while the ensemble covers a wide band — multi-resolution analysis with a quantum basis per resolution level.

3.8 Circuit copies

A single circuit reads all of its Fourier amplitudes out through one expectation value, so every amplitude is a function of the same few angles: changing one frequency’s amplitude moves the

Algorithm 1 Spectrum-Matched Circuit Design (SMCD)

Require: PDE operator $\mathcal{L}$, forcing f , boundary data g , domain Ω_x , tolerance ε , qubit budget $n_{\max}$

Ensure: encoding generators & scalings $\{\omega\}$, depth L , qubits n , entangler, observable, ansatz form

- 1: **Symbol:** $\sigma(k) \leftarrow$ Fourier symbol of $\mathcal{L}$ (linearize about a base state if nonlinear; Burgers uses the viscous-scale estimate $k_{\max} \sim 1/\sqrt{\nu}$)
 - 2: **Target:** $\hat{S} \leftarrow \{k : |\hat{f}(k)/\sigma(k)| > \varepsilon\} \cup$ boundary-lift spectral support (time-dependent: union over t , record growth rate)
 - 3: **Band:** $K \leftarrow \max |k|$ in $\hat{S}$; $\Delta \leftarrow$ min spacing in $\hat{S}$; $d \leftarrow \dim(x)$
 - 4: **Depth:** $L \leftarrow \lceil \log_3(2K/\Delta + 1) \rceil$ with ternary scalings $\omega_j = \Delta \cdot 3^{j-1}$ (Section 3.2; fall back to linear scaling if $\hat{S}$ is a dense low band)
 - 5: **Width:** $n \leftarrow d + n_{\text{cross}}$, where n_{cross} covers required mixed-frequency terms (e.g. 2-D Helmholtz needs $k_x \pm k_y \Rightarrow$ entangle the two coordinate wires)
 - 6: **if** $n > n_{\max}$ or $L > L_{\max}$ (barren-plateau depth budget) **then**
 - 7: split the band into an octave ensemble (Section 3.7)
 - 8: **end if**
 - 9: **Entangler:** ring-CZ if cross terms are needed, else none (unnecessary entanglement costs gradient variance for no benefit on a separable target)
 - 10: **Observable:** local $\langle Z_0 \rangle$ (never a global Pauli string — barren plateaus)
 - 11: **Ansatz:** hard-constrained BCs, $u = B(x) + D(x) \cdot [w \cdot Q_\theta(E_\varphi(x))]$
 - 12: **Init:** small-angle $\theta \sim \mathcal{N}(0, \sigma_0^2)$, $\sigma_0 = 0.1$; identity-block init for the encoder
 - 13: **Report:** emit a design card ($\hat{S}, \Omega, \text{coverage} = |\hat{S} \cap \Omega|/|\hat{S}|, L, n, \text{param count, predicted NTK band}$), committed alongside every experiment run
-

others. The copy model removes that coupling without changing the frequency set. It evaluates K copies of the SMCD circuit on the same encoded input, each with its own trainable angles, and mixes them with a linear head,

$$u(x) = B(x) + D(x) \left[b + \sum_{j=1}^K w_j Q_{\theta_j}(E_\varphi(x)) \right].$$

Every copy has the same encoding scalings, so the model’s frequency set is still exactly Ω and every design-card statement above applies unchanged; only the number of independent handles on the amplitudes grows. $K = 1$ is the plain SMCD circuit. On the one-dimensional problems each copy is a one-qubit, four-layer circuit, and $K = 8$ gives 91 trainable parameters. K and the learning rate were selected on tuning seeds before any held-out run (Section 4.2).

4 Measurement protocol

4.1 Explainable-AI instruments

Six instruments run at the same checkpoints ($\{0, 100, 500, 1k, 5k, 20k, \text{final}\}$ steps) for every model, problem and seed, on matched classical and hybrid families, so a measured difference cannot come from when or how each family was probed. This addresses the post-hoc storytelling that PINN explainability work is criticised for [8]: each instrument’s expected signature is stated as a claim before the runs (Section 4.3).

NTK spectroscopy. Following Faroughi and Mostajeran [3] for classical physics-informed solvers, extended here to the hybrid case: the empirical NTK on a fixed probe set, its eigenvalue spectrum, decay exponent, condition number and effective rank. The expected signature is a flatter spectrum inside the encoded band Ω than outside it (Proposition 4). NTK drift $\|\Theta_t - \Theta_0\|_F / \|\Theta_0\|_F$ tracks whether a model stays in the lazy-training regime.

Per-frequency error trajectory. The error field $u_\theta - u^*$ projected onto Fourier modes at each checkpoint, shown as a frequency-by-step map with Ω overlaid. A classical PINN gives the spectral-bias staircase; if the hybrid model’s improvement falls inside Ω , the mechanism is visible directly rather than inferred from an aggregate error.

Collocation-point attribution. Integrated Gradients of the residual loss with respect to the input coordinates, with the completeness violation and the correlation between attribution and true residual reported alongside the field, since Integrated Gradients is known to be unstable.

Fisher information and effective dimension. The empirical Fisher spectrum and the effective dimension of Abbas et al. [1], tracked through training for every family, to separate “more expressive” from “more parameters”.

Gradient variance. $\text{Var}[\partial_\theta L]$ against qubit count and depth, compared with the 2^{-n} decay expected for a global cost on a deep random circuit [7].

Layerwise probes and loss landscapes. Ridge-regression R^2 per layer for how much of u^* is linearly decodable there, and filter-normalised two-dimensional loss-landscape slices around the trained point.

4.2 Experimental setup

Problems. Six one- and two-dimensional instances on the unit domain, all with exact or high-accuracy references: Poisson, $-u'' = f$ with solution $\sin \pi x + 0.3 \sin 15\pi x$, a slow and a fast wave; the heat equation with a 15π ripple in the initial condition, a smoothing operator; viscous Burgers, nonlinear; and two-dimensional Helmholtz at $k = 4, 10, 20$. A seventh, groundwater flow under irrigated farmland, is the application of Section 7.

Model families. Every family uses the same hard boundary ansatz and training loop. The SMCD circuit (one copy, or K copies, Section 3.8); the same circuit with random encoding scalings drawn from $U(0.5, 3)$, the size-matched test of the design itself; a parallel hybrid (an MLP beside the circuit) and an octave-split ensemble (Section 3.7); an MLP (8,513 parameters on Poisson), a Fourier-feature network (129) and a frequency-matched classical model, a linear head on sine and cosine features at exactly the frequencies SMCD targets, sized to match the circuit. SMCD builds one-qubit, four-layer circuits for Poisson, heat and Burgers ($|\Omega| = 81$, 14–18 parameters) and three-qubit, one-layer circuits with a ring of CZ gates for Helmholtz ($|\Omega| = 15$). Only the frequency-matched model and the random-frequency circuit are size-matched to the circuit; the MLP and Fourier-feature network run at their standard sizes. On the groundwater problem an MLP of the circuit’s size is added.

Training. 20,000 Adam steps followed by 2,000 L-BFGS steps, five seeds per configuration. The one-copy circuit and all six problems use seeds 0–4 at a common learning rate. The eight-copy circuit was selected on the Poisson problem with seeds 0–2 from eight configurations ($K \in \{1, 2, 4, 8\}$, two learning rates), while each classical family received eight learning-rate candidates, under a selection rule fixed before any tuning run; all of its reported results are on held-out seeds 10–14, which played no part in selection. Circuits are simulated exactly by a statevector simulator cross-checked against PennyLane to 10^{-10} .

4.3 Pre-registered claims and how they are scored

Every claim about the method was written with a deciding metric and a numeric threshold in `docs/predictions*.md` and committed to git before the runs that test it; `scripts/verify_preregistration.py` checks that the committing revision is an ancestor of every such run. Claims about the eight-copy circuit were written after model selection on the tuning seeds and before any held-out run. A script (`scripts/adjudicate_predictions.py`, `adjudicate_groundwater.py`) issues every verdict from the committed run records. A claim is *confirmed* when its threshold is met and, for a comparison between two models, all five paired seeds agree (paired Wilcoxon $p \leq 0.0625$, the smallest value five pairs allow); *refuted* when the threshold is missed; *inconclusive* when the median meets the threshold but the seeds disagree. No threshold is changed after a run.

5 Results

Errors are relative L^2 against the reference solution, as the median over five seeds. Table 1 lists every pre-registered claim about the one-copy circuit and Table 3 those about the eight-copy circuit; the sections below take the evidence question by question.

Claim	Verdict	Measured
Beats the large MLP on Poisson ($\geq 2\times$)	Refuted	$0.65\times$
Beats Fourier features on Poisson ($\geq 1.3\times$)	Refuted	$0.13\times$
Matches the frequency-matched classical model on Poisson (within $1.3\times$)	Refuted	$5.11\times$
Heat: no model beats the MLP by more than 5%	Confirmed	
Helmholtz: the gain over the MLP grows with k	Confirmed	
Designed beats random frequencies, Poisson ($\geq 1.5\times$)	Inconclusive	$1.84\times$
Designed beats random frequencies, heat ($\geq 1.5\times$)	Confirmed	$6.52\times$
NTK flatter inside Ω than outside, Poisson	Insufficient data	
NTK flatter inside Ω than outside, Helmholtz $k=10$	Refuted	gap -8.4
Gain over the MLP lies inside Ω , Poisson ($\geq 70\%$)	Confirmed	99.9%
Gain over the MLP lies inside Ω , Helmholtz $k=10$	Refuted	
Higher coverage gives lower error ($\rho \leq -0.7$)	Refuted	
Gradient variance stays above the barren-plateau line	Confirmed	slope 0.31, R^2 0.001
Leaves the lazy-training regime earlier than classical models	Refuted	1.01 vs 4.3e+06
Encoder drift stays below 0.2, so Ω stays valid	Confirmed	0.13
Confirmed	6 of 15	

Table 1: Pre-registered claims about the one-copy SMCD circuit, seeds 0–4, as the adjudication script scores them. Ratios are baseline error over circuit error, except the frequency-matched comparison (circuit over baseline) and drift (hybrid vs classical).

5.1 The circuit realises its design

For every one-dimensional problem, the SMCD circuit was evaluated at 40 random parameter settings and its output Fourier-transformed. The largest amplitude outside the designed set Ω , relative to the largest overall, is at most 3.8×10^{-15} , i.e. floating-point zero (Figure 2), and the test suite checks the same property on every build. The encoder that feeds the circuit could in principle move Ω during training, but it barely does: the drift $\|A - I\|_F$ at the end of training is 0.13 for one copy and 0.10 for eight, under the 0.2 at which the coverage computed at initialisation would stop describing the trained model. The design card is therefore a statement about the trained model, not only its starting point.

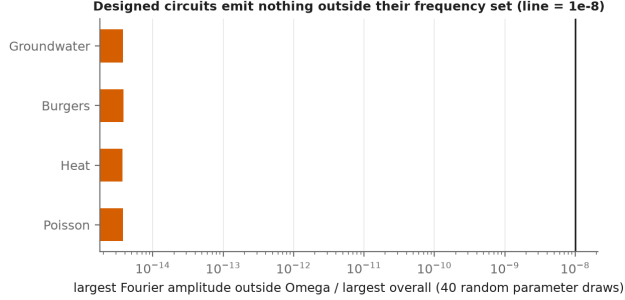


Figure 2: Largest Fourier amplitude outside Ω for each problem’s SMCD circuit, over 40 random parameter draws (line: 10^{-8}). The two-dimensional Helmholtz designs are not covered by this one-dimensional check.

5.2 Accuracy across six problems

Table 2 and Figure 3 give every family’s error on every problem with one copy of the circuit. Three regimes appear.

Family	Poisson	Heat	Burgers	Helmholtz		
				$k=4$	$k=10$	$k=20$
MLP	1.08	0.305	0.014	5.0×10^{-5}	1.00	1.00
Fourier features	0.21	0.39	0.33	0.92	1.00	1.00
Frequency-matched	0.33	0.33	0.31	0.95	1.00	1.00
SMCD circuit	1.68	0.42	1.14	1.59	1.09	1.00
Random frequencies	3.08	2.74	1.06	0.95	1.01	1.01
Parallel hybrid	0.35	0.76	0.26	0.26	1.00	1.00
Octave ensemble	0.75	0.68	0.85	1.00	1.00	1.00

Table 2: Median relative L^2 error, one copy of each circuit, seeds 0–4 (two to four seeds for six Helmholtz $k=10/20$ cells). Bold: best per problem. Predicting $u \equiv 0$ scores exactly 1.

Where a slow and a fast wave mix (Poisson, Burgers, Helmholtz $k=4$), classical models win against one copy. On Poisson the circuit’s error (1.68) is higher than the MLP’s (1.08), the Fourier-feature network’s (0.21) and the frequency-matched model’s (0.33); all three comparisons are refuted. Section 5.5 traces this to coupled amplitudes and removes it with copies.

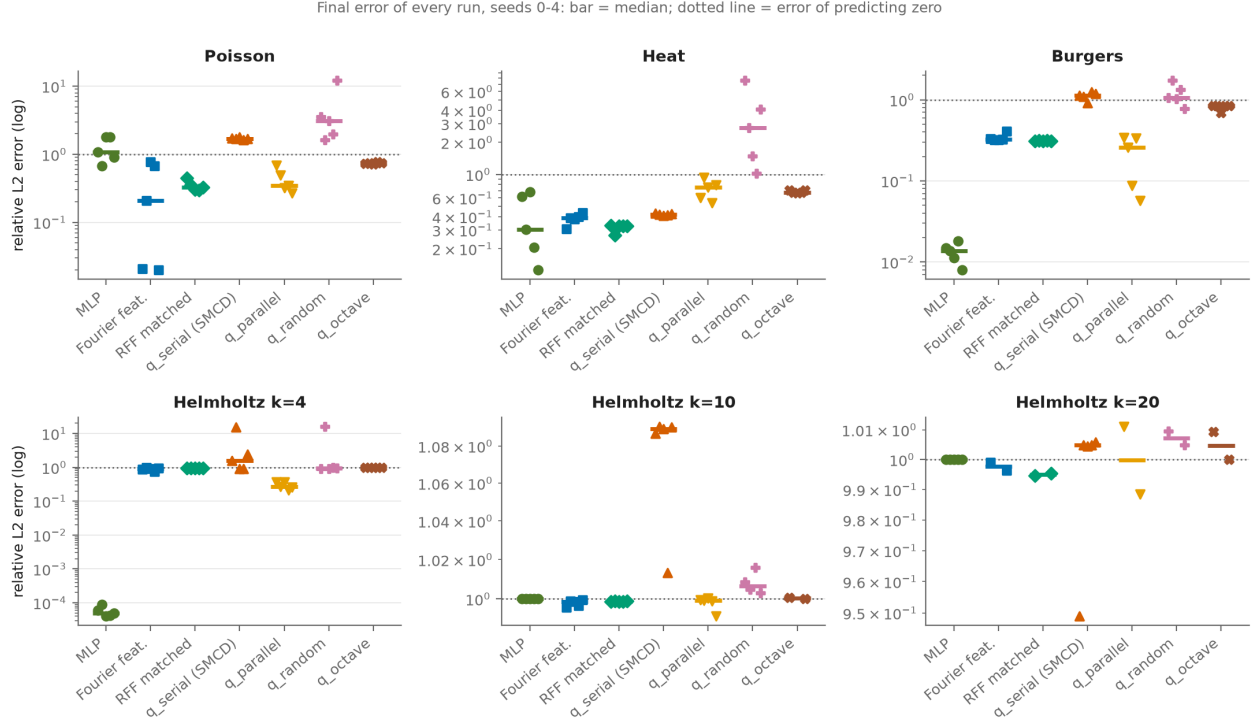


Figure 3: Final relative L^2 error of every run, per problem and family.

On a smoothing operator (heat), nothing beats a plain MLP. SMCD predicts this before training (Proposition 2): diffusion removes the high-frequency content a circuit could supply, and the design card’s predicted benefit is 2.5%. Every family is worse than the MLP, the circuit by 38% and the random-frequency circuit by 9 $\times$, so the claim that no family beats it by more than 5% is confirmed.

At high wavenumber (Helmholtz $k=10, 20$), every model fails. Every family’s median error is 0.995–1.09, the score of predicting zero. The pre-registered trend that the circuit’s error ratio against the MLP improves with k is confirmed ($3 \times 10^{-5} \rightarrow 0.92 \rightarrow 0.99$), and the circuit is within 1.3 $\times$ of the frequency-matched model at both wavenumbers, but both are parity among models that did not learn the solution. A plausible cause, not tested here: at these k the forcing term is about 1% of the Laplacian term, so $u \equiv 0$ leaves a small residual. SMCD’s own predicted benefit is zero at $k = 4$ and $k = 10$.

Cost. On a classical simulator a circuit run takes a median of about 2,900s, against 330–590s for the classical families, with the same number of optimiser steps.

5.3 Designed frequencies beat random ones

The sharpest test of SMCD holds the circuit, its size and its training fixed and changes only the encoding scalings, from SMCD’s to random ones (Figure 4). Random scalings drawn from $U(0.5, 3)$ reach at most about 12 radians per unit in four layers, short of the $15\pi \approx 47$ wave, so the comparison tests whether reaching the needed band matters. Wherever the circuit learns the solution, it does:

- **Heat, one copy:** $6.5\times$ lower error on all five seeds (95% bootstrap interval $2.5\text{--}17.7\times$, Section 6). The designed circuit is also the most consistent model in the matrix (worst over best seed within 4%), while its random twin is the least ($7.4\times$).
- **Poisson, eight copies:** $236\times$ (0.0037 against 0.87), all five held-out seeds.
- **Groundwater, eight copies:** $9.9\times$ (0.09% against 0.92%), all five held-out seeds (Section 7).

On Poisson with one copy the median ratio is $1.84\times$, but one seed of five disagrees, so the claim is inconclusive; this is the case where one copy does not learn the solution at all.

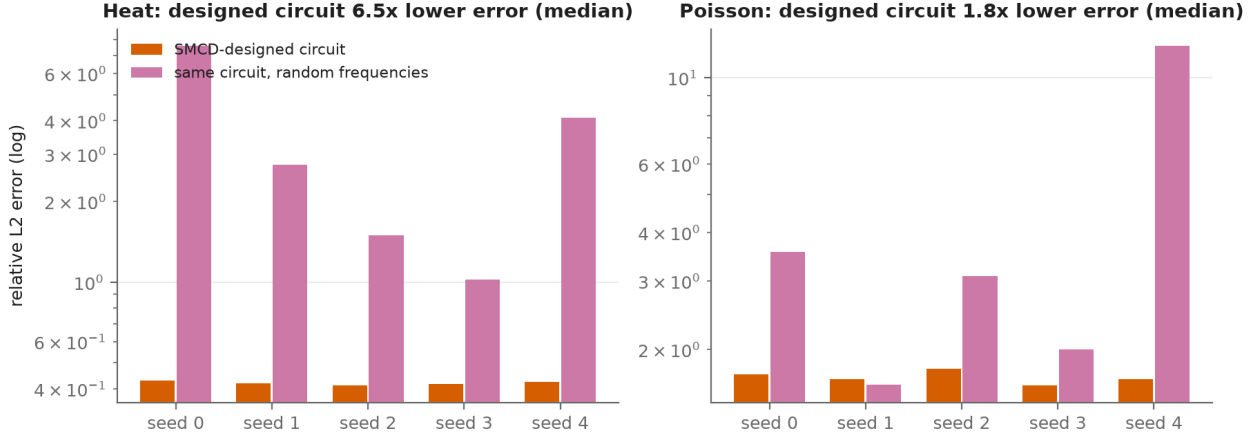


Figure 4: One copy of the circuit with SMCD’s scalings against the same circuit with random scalings, every seed, heat (left) and Poisson (right).

5.4 The mechanism in the frequency domain

The gain lands on the designed frequencies. Projecting the error onto Fourier modes and accumulating the improvement over the MLP frequency by frequency, 99.9% of the circuit’s improvement on Poisson lies inside Ω with one copy, 99.9% with eight, and 99.4% on groundwater (the bar was 70%). On Helmholtz $k=10$ the circuit improves on the MLP at no frequency at all, so there is nothing to attribute (Figure 5).

The NTK is flatter inside Ω once the model can express it. Proposition 4 predicts a flatter NTK spectrum inside the encoded band. With eight copies on Poisson the decay exponent inside Ω is flatter than outside by 15.2 (the bar was 0.5). With one copy the comparison cannot be made on Poisson: the model has 14 parameters, so its NTK has at most 14 non-zero eigenvalues and none beyond index $|\Omega| = 81$. On Helmholtz $k=10$, where every model fails, the spectrum is steeper inside Ω (gap -8.4) (Figure 6).

Hybrid models stay closer to their initial kernel. The prediction that hybrids leave the lazy-training regime earlier is reversed: at 5,000 steps the median relative NTK drift is 1.0 for the one-copy hybrid and 4.3×10^6 for the classical models (2.5 against 9.7×10^4 with eight copies). The circuits also use few directions in parameter space: an NTK effective rank of 1.4 on Poisson and 1.1 on heat, and a Fisher effective dimension of about 4, flat through training, where the MLP’s

grows from about 4 to 12 (Poisson) and 5 to 50 (heat). The size-matched classical model looks like the circuit (effective rank 2.0 and 1.9), so this is mostly an effect of size, not of being quantum (Figure 22).

Coverage alone does not predict error. SMCD’s design card reports the fraction of the target spectrum inside Ω . Sweeping the coverage target on Poisson builds only three distinct circuits (a target of 0.8, 0.9 or 1.0 yields the same one), so coverage there is confounded with parameter count, and at the full training budget the correlation between coverage and error is $\rho = +0.29$, the wrong sign for the pre-registered $\rho \leq -0.7$. On Helmholtz $k=10$ every target builds the same circuit, which scores about 1 throughout. Covering the band is necessary for the design to help (Section 5.3) but not sufficient for a single circuit to train well.

No measurable barren plateau at this scale. Gradient variance fitted against qubit count (4 and 6 qubits, three seeds) has slope 0.31, under the $\log 2 = 0.69$ bar, but the fit explains almost none of the variation ($R^2 = 0.001$): no decay is detectable between 4 and 6 qubits. Eight qubits did not fit in the simulator’s memory.

5.5 Why one copy is not enough

One copy learns the hard frequency and loses the easy one. The Poisson solution is $\sin \pi x + 0.3 \sin 15\pi x$. Tracking the error at those two frequencies (Figure 7), one copy cuts its 15π error to a median 10% of its starting value by step 1,000 (103% for the Fourier-feature network, 86% for the MLP, 51% for the frequency-matched model). Its π error falls to about 30% by step 5,000 and then *rises* to $1.9\text{--}2.0\times$ its starting value by step 20,000, on all five seeds, while the frequency-matched model reduces both. The residual loss weights each Fourier mode of the error by ω^2 , so the 15π mode carries 225 times the weight of the π mode. A linear head on fixed sine features sets the two amplitudes independently; a single circuit cannot, because all 81 of its amplitudes are functions of the same ten angles read through one expectation value. Our reading, supported by these data but not proven, is that the optimiser buys the heavily weighted amplitude at the cost of the lightly weighted one. The loss agrees: it stalls near 3.3×10^4 from step 1,000 while the error keeps moving, and the seeds agree closely (worst over best $1.1\times$) because they converge to the same trade-off.

Copies decouple the amplitudes. Section 3.8’s copies keep Ω and give the head an independent handle on each copy. On the tuning seeds the error falls with the number of copies, from 1.67 with one copy to 0.003 with eight at the higher learning rate (Figure 8); eight copies were selected.

Eight copies on held-out seeds. On seeds 10–14 (Table 3, Figure 9) the eight-copy circuit reaches a median error of 0.0037, with every seed below 0.02. It beats the Fourier-feature network by three orders of magnitude and the same circuit with random frequencies by $236\times$. The comparison with the MLP is inconclusive: at its tuned learning rate the MLP failed on three of five seeds (1.2–3.6) and reached 0.003–0.012 on the other two. The classical learning rates chosen on the tuning seeds did not transfer well (the Fourier-feature network went from 0.19 to 5.0), which inflates both of those ratios. **The frequency-matched classical model, sized to the eight-copy circuit (83 parameters), is $3.3\times$ more accurate (0.0011):** on Poisson, handing a classical model the frequencies SMCD reads off the equation works better than the circuit.

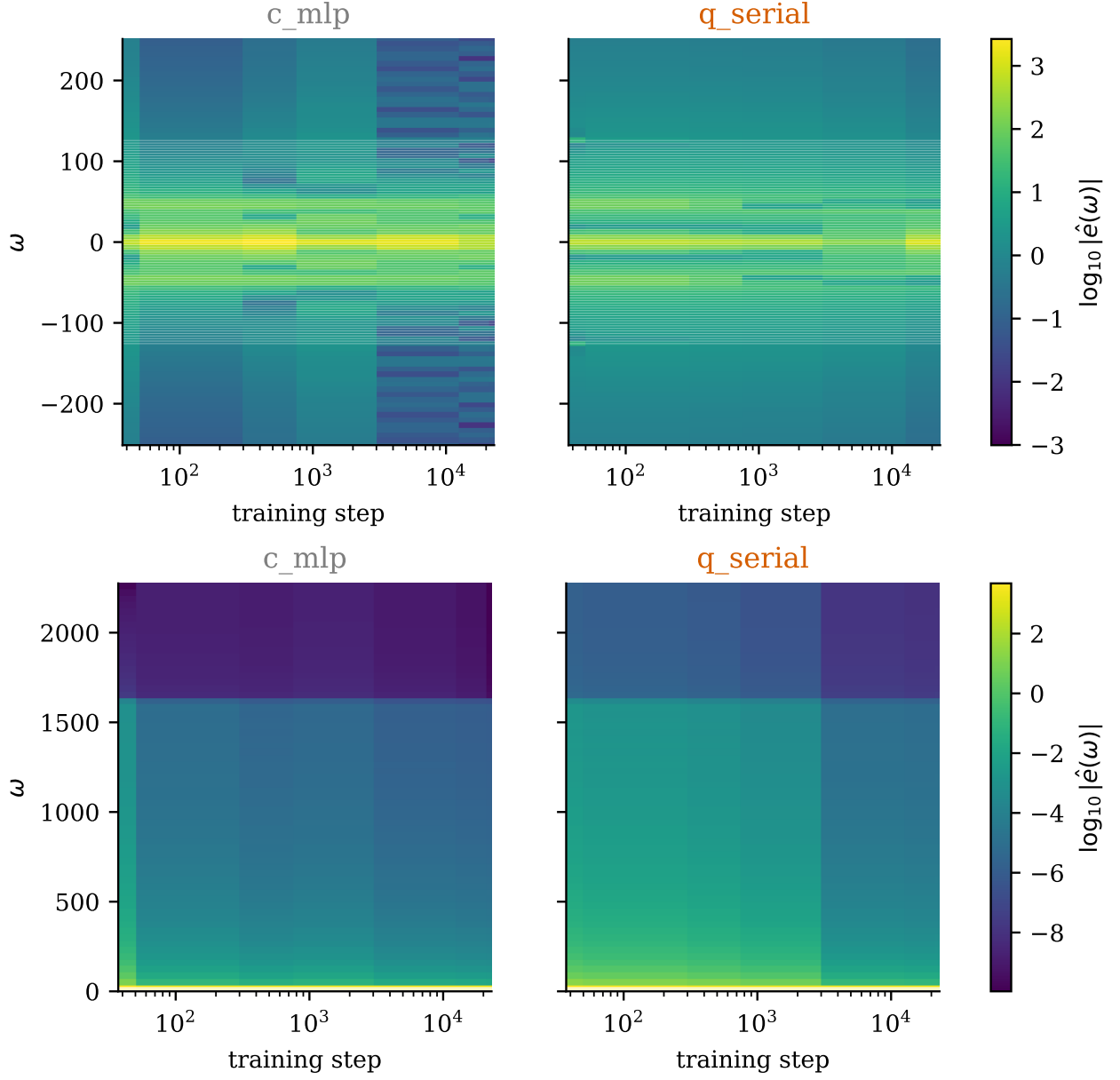


Figure 5: Per-frequency error $|\hat{e}(\omega)|$ over training, MLP (left, **c_mlp**) against one copy of the SMCD circuit (right, **q_serial**), seed 0, with Ω overlaid as thin white lines. Top: Poisson, two-sided spectrum within $\pm 2 \max |\Omega|$. Bottom: Helmholtz $k=10$, radially binned; all of Ω ($|\omega| \leq 19$) falls in the lowest bin.

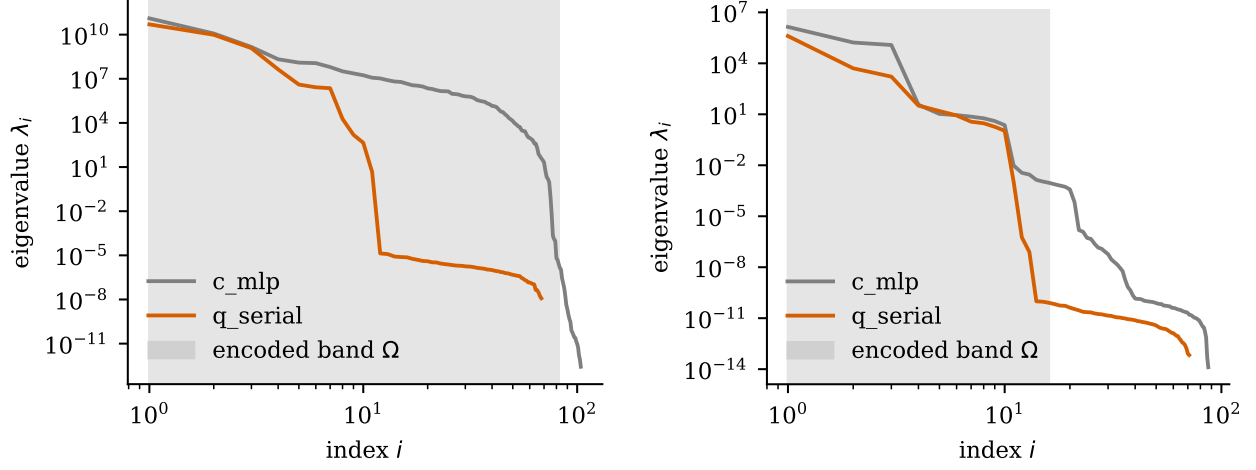


Figure 6: Residual NTK eigenvalue spectra at the final checkpoint, MLP (c_mlp) against one copy of the SMCD circuit (q_serial), seed 0, with the top- $|\Omega|$ index range shaded. Left: Poisson ($|\Omega| = 81$ exceeds the circuit's 14 parameters). Right: Helmholtz $k=10$ ($|\Omega| = 15$).

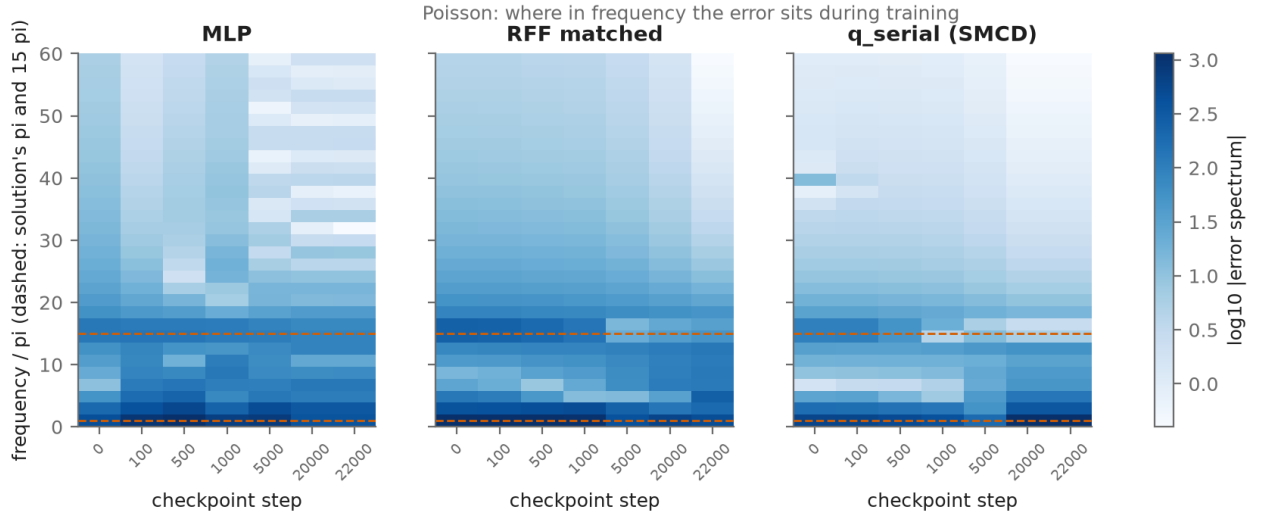


Figure 7: Poisson, one copy: magnitude of the error spectrum at each checkpoint (seed 0), with the error at π and 15π tracked over training.

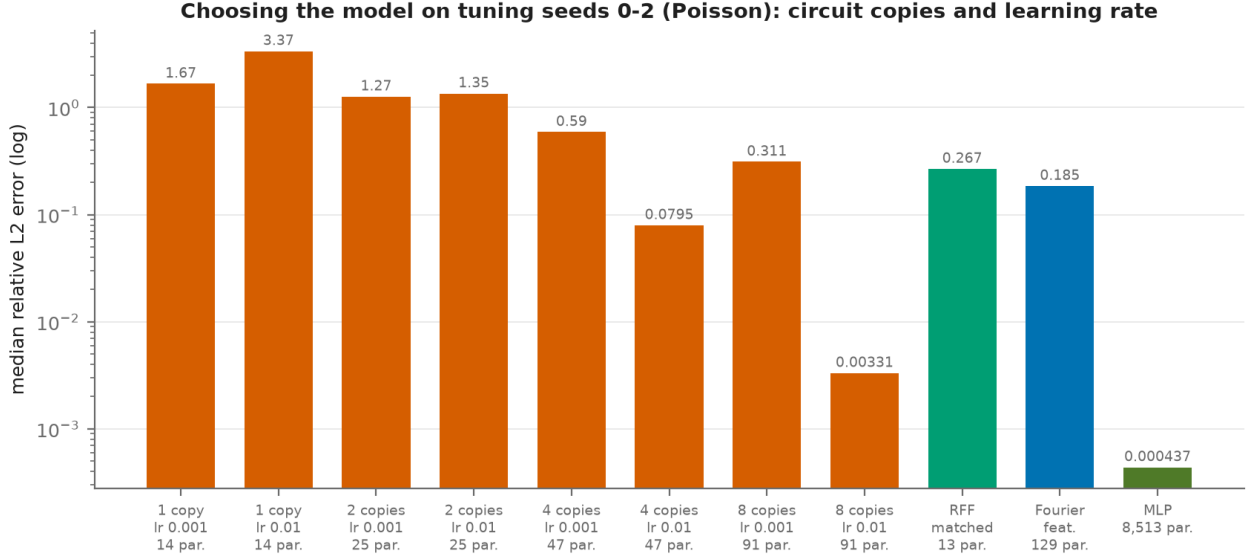


Figure 8: Model selection on Poisson, tuning seeds 0–2: median error for 1, 2, 4 and 8 copies at two learning rates, beside the best learning rate for each classical model.

Claim	Verdict	Measured
Beats the large MLP on Poisson ($\geq 2\times$)	Inconclusive	337.79 $\times$
Beats Fourier features on Poisson ($\geq 1.3\times$)	Confirmed	1359.81 $\times$
Matches the frequency-matched classical model on Poisson (within 1.3 $\times$)	Refuted	3.30 $\times$
Designed beats random frequencies, Poisson ($\geq 1.5\times$)	Confirmed	236.08 $\times$
NTK flatter inside Ω than outside, Poisson	Confirmed	gap 15.2
Gain over the MLP lies inside Ω , Poisson ($\geq 70\%$)	Confirmed	99.9%
Leaves the lazy-training regime earlier than classical models	Refuted	2.53 vs 9.7e+04
Encoder drift stays below 0.2, so Ω stays valid	Confirmed	0.10
Confirmed	5 of 8	

Table 3: Pre-registered claims about the eight-copy SMCD circuit on Poisson, held-out seeds 10–14. The baselines are sized to the eight-copy circuit where the claim names one.

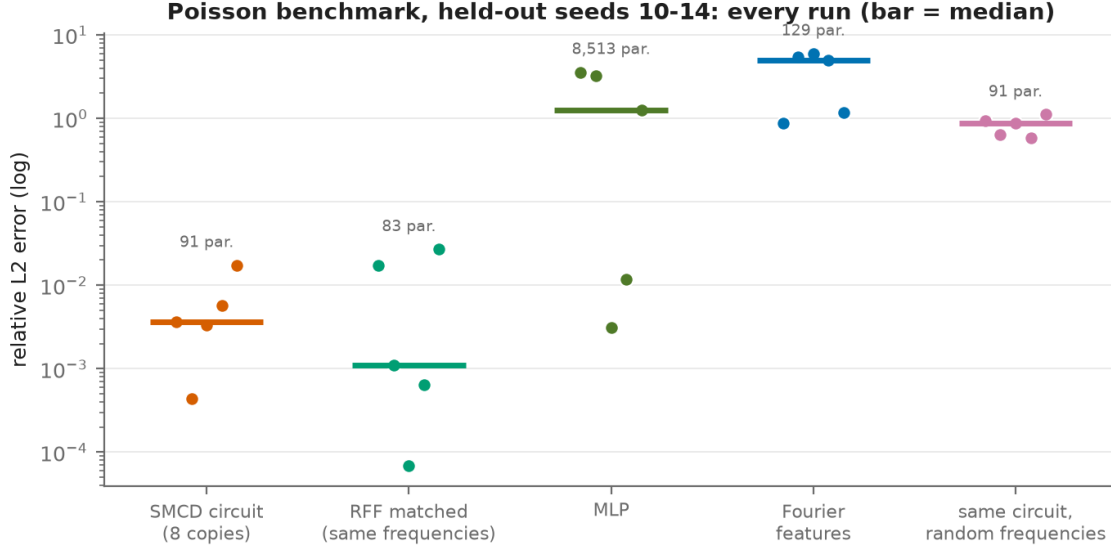


Figure 9: Poisson, held-out seeds 10–14: every run of the eight-copy circuit and the baselines (bar: median).

6 Robustness of the verdicts

Every verdict comes from one pre-registered rule applied to five paired seeds. This section asks how much the verdicts on the six-problem matrix depend on that rule and on individual seeds. All of it is computed from the committed runs (`scripts/report/fig_validation.py`, record in `results/report/validation.json`); no model was retrained.

Bootstrap intervals. Figure 10 resamples the five paired seeds 5,000 times and gives a 95% interval for each pre-registered error ratio. The heat design-versus-random result is robust: the designed circuit’s error is $6.5\times$ lower, with the whole interval ($2.5\times$ to $17.7\times$) above the $1.5\times$ bar. The one-copy Poisson comparisons against classical models fail decisively rather than narrowly: against the MLP the interval is 0.40 to 1.10 (bar $2\times$), against Fourier features 0.01 to 0.43 , and against the frequency-matched model 3.9 to 5.8 , entirely outside its equivalence band. The one-copy Poisson design-versus-random ratio (1.84 , interval 0.97 to 7.2) straddles its threshold, as an inconclusive verdict should.

Other tests. With five paired seeds the smallest two-sided Wilcoxon p is $1/16 = 0.0625$ however large the effect, so the confirmation rule can only be met by five seeds of five agreeing. Figure 11 applies three alternatives to the same pairs. They agree with every verdict and show what the Wilcoxon floor hides: on the heat design-versus-random comparison a paired t -test on log error gives $p = 0.006$ and a Mann–Whitney test $p = 0.008$.

Leave one seed out. If one seed decided a verdict, dropping it would move the ratio across its threshold. Figure 12 repeats each ratio five times with one seed removed. The heat design-versus-random ratio clears its bar in all five, the one-copy Poisson comparisons against classical models miss theirs in all five, and the one-copy Poisson design-versus-random ratio clears it in four of five.

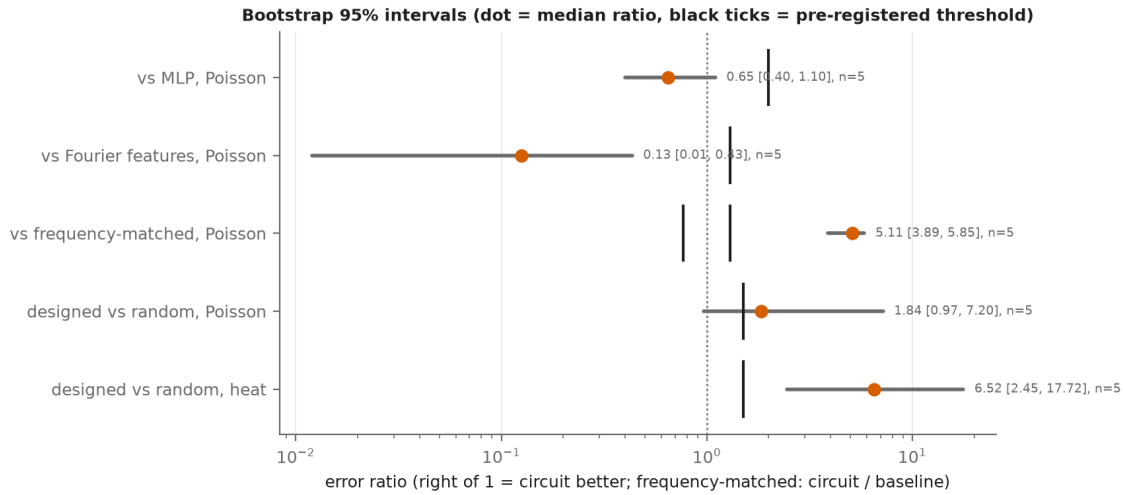


Figure 10: Bootstrap 95% intervals of the pre-registered error ratios, one copy of the circuit (5 paired seeds, 5,000 resamples). Right of 1 means the circuit is better; for the frequency-matched comparison the ratio is circuit over baseline and must lie between the two ticks.

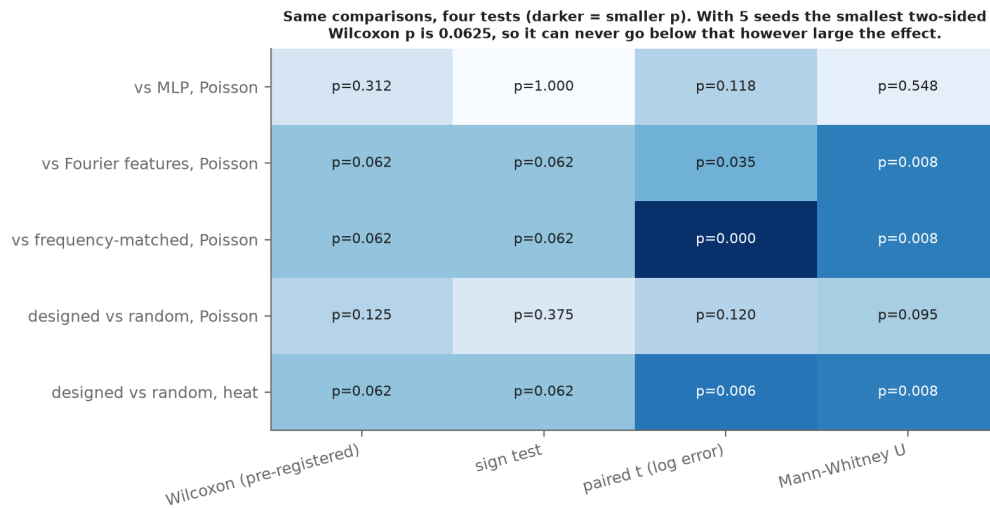


Figure 11: The pre-registered Wilcoxon test beside three alternatives, on the same paired seeds. Darker cells have smaller p .

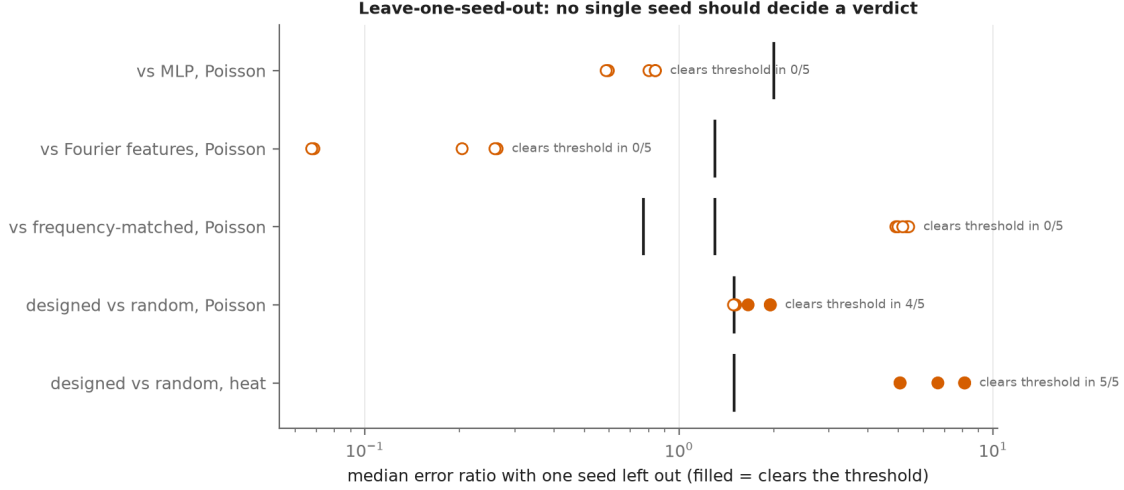


Figure 12: Each ratio recomputed with one seed left out (five points per row). Filled points clear the pre-registered threshold.

7 Application: waterlogging under canal-irrigated farmland

SMCD needs a solution's frequencies before training. Real problems where those are known in advance come from regular structure: evenly spaced sources, periodic geometry. This section applies the method to one such problem, with its claims pre-registered like all others (Section 4.3).

7.1 The problem

Two rivers $L = 2$ km apart bound a confined aquifer beneath irrigated farmland. Water enters from background recharge R_0 (rain and irrigation return flow) and from six canals running parallel to the rivers at equal spacing, each leaking q m³/day per metre of its length into a strip of width σ . Dupuit's steady one-dimensional equation for the hydraulic head h (water-table height above the river datum) is

$$T h''(X) = -R(X), \quad R(X) = R_0 + \sum_{j=1}^6 \frac{q}{\sigma \sqrt{2\pi}} e^{-(X-c_j)^2/2\sigma^2}, \quad h(0) = h_A, \quad h(L) = h_B, \quad (1)$$

with $c_j = (j - \frac{1}{2})L/6$. The scenario uses values within textbook ranges [4, 13], not calibrated to one site: transmissivity $T = 100$ m²/day (silty sand), $R_0 = 0.4$ mm/day, $q = 0.4$ m²/day, $\sigma = 30$ m, $h_A = 1$ m, $h_B = 0$. On $x = X/L$ this is $-u'' = f$, the Poisson operator with non-zero boundary values, so every model and instrument applies unchanged. The canal spacing fixes the frequencies, harmonics of $2\pi \cdot 6$ in x , and SMCD designs a one-qubit, four-layer circuit for them.

Because $\sigma(z\Phi(z) + \phi(z))$ with $z = (X - c)/\sigma$ has second derivative $\phi(z)/\sigma$, the head has a closed form. An independent second-order finite-difference solver agrees with it to 5×10^{-8} m and converges to it at measured order 2.00 as the grid is refined from 100 to 6,400 intervals (Figure 14), which checks the closed form, the forcing and the boundary treatment together.

7.2 The practical question

A field is waterlogged where the water table rises within 1.5 m of the land surface, into the crop root zone. With the surface 9 m above the river datum, the water table peaks at 8.53 m and 713 m

of the 2 km field is waterlogged (Figure 13). The solution answers the planning questions a farmer or irrigation engineer would ask (Figure 14):

- **Lining the canals.** Cutting canal seepage by 20% dries the whole field; a 10% cut reduces the waterlogged length from 713 m to 494 m.
- **Soil.** The same canals waterlog 1,778 m on clay loam ($T = 25 \text{ m}^2/\text{day}$) and none of the field on fine sand ($T \geq 150$).
- **Spacing.** Concentrating the same total seepage into two canals waterlogs 1,066 m and raises 84 cm mounds under each canal; spreading it over twelve reduces the mounds to 1 cm.

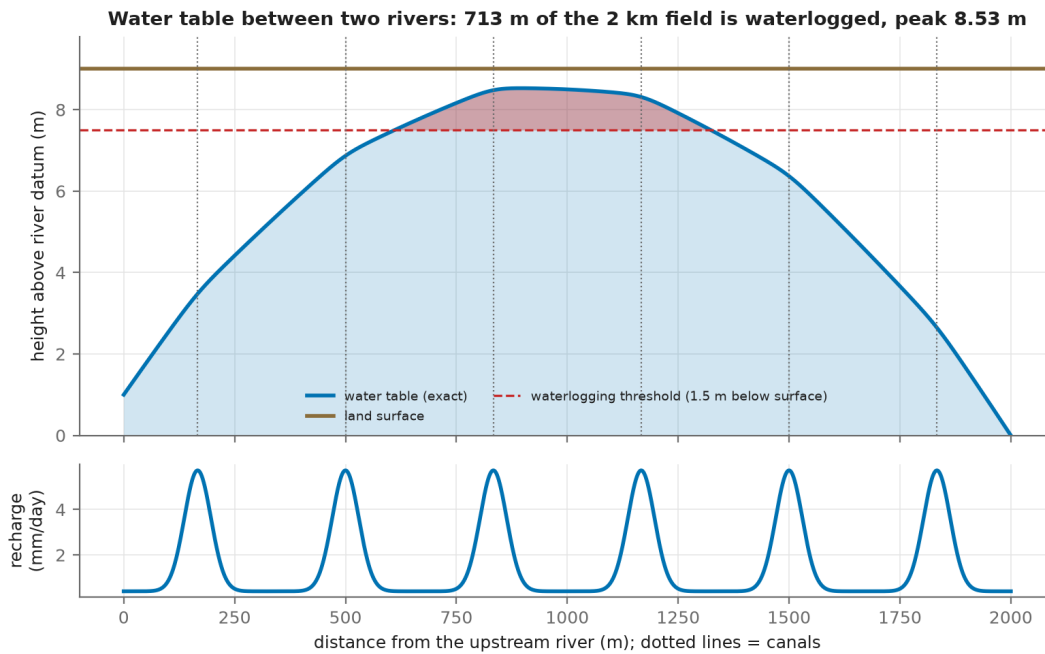


Figure 13: Exact water table between the two rivers (top) and the recharge that drives it (bottom). Red: the 713 m of field where the water table is within 1.5 m of the surface.

7.3 Can the trained models answer it?

Five claims were pre-registered for the SMCD circuit and tested on held-out seeds 10–14 at the full training budget, with one copy and with eight, each against baselines of its own size (Table 4).

Eight copies meet all five claims. The median error is 0.09%, with every seed below 0.5%. That is 18× more accurate than an MLP with the same 91 parameters, 9.9× more accurate than the same circuit with random frequencies (both on every paired seed), and more accurate than the 8,513-parameter MLP. 99.4% of its improvement over the MLP lies inside Ω , and it places the waterlogged length at 714 m against the exact 713 m (Figure 15). Unlike on Poisson, the classical model handed the same frequencies does not match it here (22.9% error, no waterlogging predicted). A plausible reason, not tested here, is that the solution’s broad mound is poorly represented by a linear head on fixed sine features, while the circuit’s trainable encoder can adjust its frequencies.

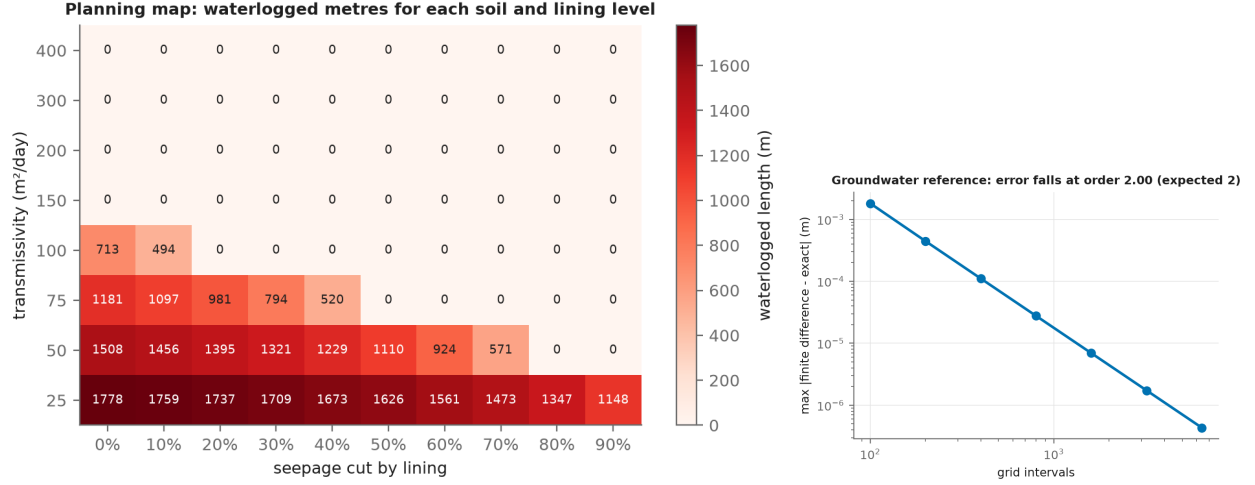


Figure 14: Left: waterlogged length for each combination of soil (transmissivity) and canal lining (seepage cut). Right: finite-difference error against the closed form as the grid refines.

Claim	1 copy	8 copies
Error below 1%	Refuted (52.76%)	Confirmed (0.09%)
Beats a classical network of the same size ($\geq 1.3\times$)	Refuted (0.09 $\times$)	Confirmed (18.23 $\times$)
Beats the same circuit with random frequencies ($\geq 1.5\times$)	Refuted (0.11 $\times$)	Confirmed (9.91 $\times$)
Gain over the MLP lies inside Ω ($\geq 70\%$)	Confirmed (99.4%)	Confirmed (99.4%)
Waterlogged length within 5% of the exact 713 m	Refuted (0 m)	Confirmed (714 m)
Confirmed	1 of 5	5 of 5

Model	parameters	median error	waterlogged (m)
SMCD circuit, 8 copies	91	0.09%	714
random frequencies, 8 copies	91	0.92%	738
MLP, size of 8 copies	91	1.68%	672
frequency-matched, size of 8 copies	83	22.93%	0
MLP, large	8,513	1.94%	707
SMCD circuit, 1 copy	14	52.76%	0
random frequencies, 1 copy	14	5.96%	535
MLP, size of 1 copy	13	4.96%	840
frequency-matched, size of 1 copy	13	15.76%	0

Table 4: Groundwater: pre-registered claims for the SMCD circuit with one and eight copies (top) and every model’s median relative L^2 error and waterlogged length (bottom), held-out seeds 10–14. Exact waterlogged length: 713 m.

One copy meets one of five. Its improvement over the MLP still lies on the designed frequencies (99.4%), but it does not learn the water table: its median profile peaks near 4 m instead of 8.5 m, so it predicts no waterlogging (Figure 16). A classical network of its size (13 parameters) reaches 5.0% and the random-frequency circuit 6.0%. This is the Poisson trade-off of Section 5.5 again: the solution is dominated by a broad mound, the slow component a single circuit gives up to fit the canal ripple.

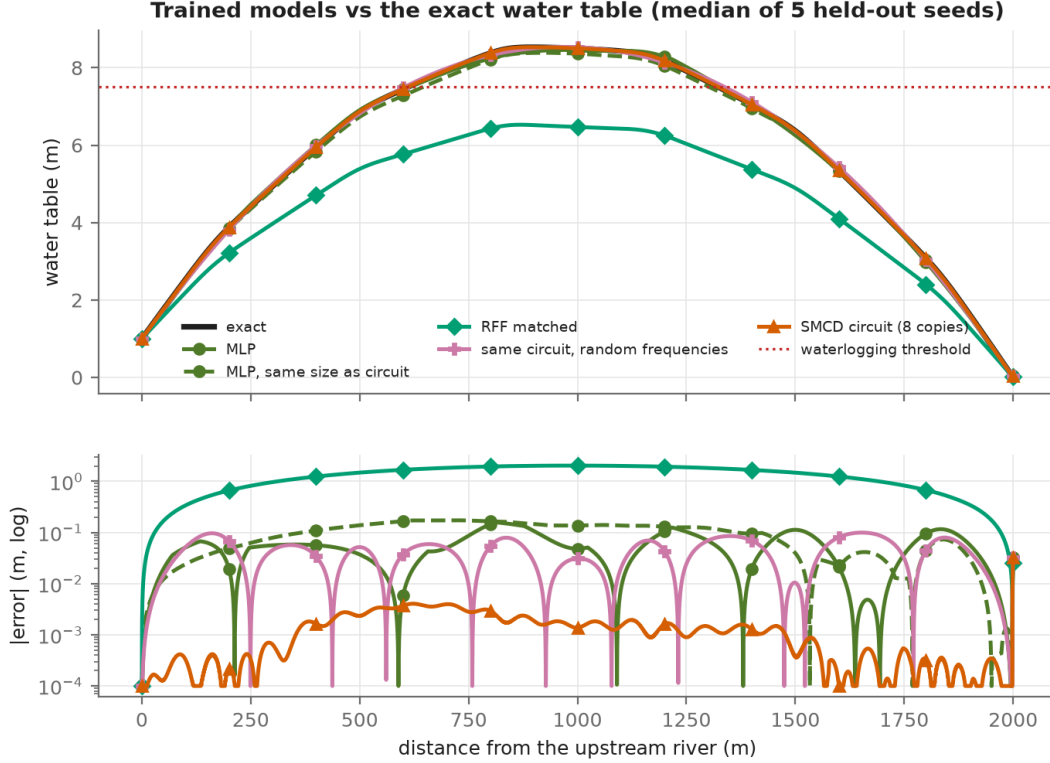


Figure 15: Eight copies of the SMCD circuit and the baselines of its size on the groundwater problem (median of seeds 10–14). Bottom: absolute error along the field; the circuit stays within a few millimetres.

8 Limitations

1. **Simulation only.** Every circuit is evaluated on a statevector simulator, cross-checked against PennyLane to 10^{-10} ; nothing here ran on quantum hardware. Shot and depolarising noise are modelled by analytic surrogates, whose agreement with a density-matrix simulation is measured on a representative subset rather than assumed.
2. **Small circuits.** On the one-dimensional problems each circuit is one qubit without entanglement, a model a classical computer evaluates cheaply; the Helmholtz circuits use three qubits. Simulation cost capped the experiments at 6–8 qubits, though SMCD’s formulas apply unchanged at larger n . No claim of quantum computational advantage is made or tested.
3. **Statistical power.** Five seeds per configuration give a smallest two-sided Wilcoxon p of

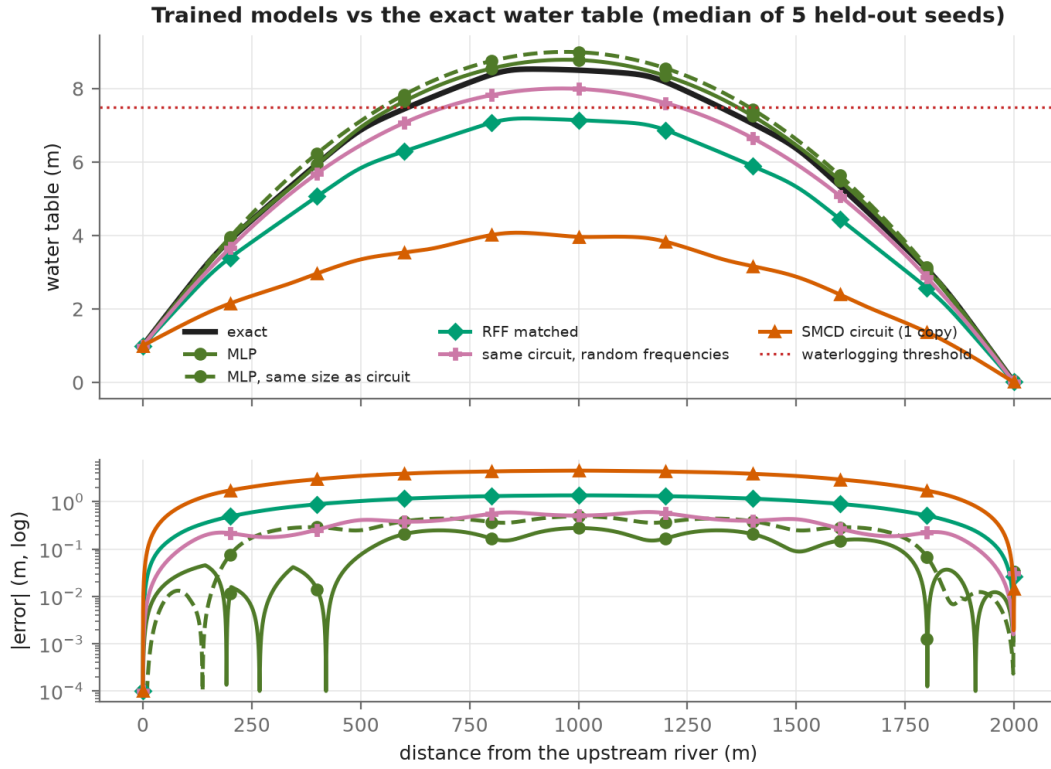


Figure 16: One copy of the SMCD circuit and the baselines of its size (median of seeds 10–14). Dotted: the waterlogging threshold.

0.0625, so “confirmed” means all five seeds agree; bootstrap intervals and three alternative tests (Section 6) support each verdict but do not raise the sample size.

4. **Where the method wins, and where it does not.** The eight-copy circuit’s strongest result is one applied problem. On Poisson a classical model given the same frequencies remains $3.3\times$ more accurate, and at high Helmholtz wavenumbers no model learns the solution, so parity there says nothing about capability.
5. **Baseline sizes and tuning.** In the six-problem matrix only the frequency-matched model and the random-frequency circuit are size-matched to the circuit; the MLP and Fourier-feature network run at standard sizes. The eight-copy model was selected from a small grid of copy counts and learning rates, while each classical family received a learning-rate search only, and those learning rates did not transfer reliably to held-out seeds.
6. **Coverage.** Achieved coverage does not predict error on its own (Section 5.4); the coverage sweep builds few distinct circuits, which confounds coverage with parameter count.
7. **The affine encoder.** The classical encoder is restricted to an affine map so that the circuit’s frequency set, and hence coverage, is well-defined; a nonlinear encoder would make Ω input-dependent. This restricts the hybrid model.
8. **The predicted-benefit number is a heuristic.** The design card’s a-priori benefit estimate is not a proven bound. It predicted 2.5% on heat and zero on Helmholtz $k=4, 10$; the circuit beat the MLP on none of them.

9 Conclusion

Spectrum-Matched Circuit Design turns the choice of a quantum circuit for a PDE into arithmetic: the operator’s Fourier symbol and the forcing give the frequencies the solution needs, and the encoding scalings, depth, qubit count and entangler follow by formula. The circuit it builds produces exactly the designed frequencies, and those frequencies matter: against the same circuit with random ones, the designed circuit is $6.5\times$ more accurate on the heat equation, $236\times$ on Poisson and $9.9\times$ on groundwater, on every seed.

The explainability instruments showed what a single circuit cannot do. All of its Fourier amplitudes depend on the same few angles, so on a solution that mixes a slow and a fast wave the optimiser trades one for the other. Mixing eight copies of the designed circuit through a linear head keeps the frequency set and frees the amplitudes. On waterlogging of farmland from leaking canals, that model reaches 0.09% error, $18\times$ more accurate than a classical network of the same size, and answers the planning question to within 1 m.

The results also mark the method’s limits. On a smoothing operator no model beats a plain MLP, as SMCD predicts before training; on Poisson a classical model handed the same frequencies is $3.3\times$ more accurate than the circuit; and at high Helmholtz wavenumbers every model fails. The most direct next steps are running the designed circuits on quantum hardware, extending SMCD to problems of higher dimension, and understanding when a trainable encoder lets the circuit beat fixed classical features at the same frequencies, as it does on the groundwater problem.

10 Reproducibility

- **Code and data.** `src/qapinn/` holds the PDEs, models, SMCD, training loop and instruments; `results/runs/` holds every run’s metrics and instrument records, from which every figure and verdict is computed without reloading a trained model. `results/README.md` maps run IDs to experiments.
- **Hardware:** NVIDIA RTX 4090 Laptop GPU, 32 logical CPU cores (`provenance.json`, recorded per run); one machine for every run.
- **Software** (`docs/ENV_RESOLVED.md`, pinned via uv): Python 3.12.13, torch 2.11.0+cu128 (CUDA 12.8), pennyLane 0.45.1, numpy 2.4.4, scipy 1.18.0.
- **Determinism:** `qapinn.seeding.set_global_seed` seeds Python, NumPy and Torch from one integer and enables deterministic algorithms; an explicit generator is threaded through every sampling call, so two runs at the same seed on the same device produce bit-identical metrics. A CPU and a CUDA run of the same configuration round differently (by 10^{-15} in relative L^2 on a checked run).
- **Seeds.** Six-problem matrix: 0–4. Coverage sweep: 0–2. Barren-plateau sweep: 0–2 at 4 and 6 qubits. Model selection for circuit copies: 0–2. Held-out evaluation of the eight-copy circuit and of all groundwater models: 10–14.
- **Commands** (`docs/REPRODUCE.md`). `tasks.py test` runs the test suite in about 4 minutes on CPU; `tasks.py figures` redraws every figure from the committed runs in about 30s; `tasks.py repro-quick` trains all six problems and seven families at reduced size on CPU in about 9 minutes; `tasks.py repro-all` re-derives every figure and verdict from the committed runs in about a minute, while retraining every run from scratch takes about 67 hours on the machine above.

References

- [1] Amira Abbas, David Sutter, Christa Zoufal, Aurelien Lucchi, Alessio Figalli, and Stefan Woerner. The power of quantum neural networks. *Nature Computational Science*, 1:403–409, 2021.
- [2] Ziv Chen, Gal G. Shaviner, Hemanth Chandravamsi, Shimon Pisnoy, Steven H. Frankel, and Uzi Pereg. Quantum physics-informed neural networks for Maxwell’s equations: circuit design, “black hole” barren plateaus mitigation, and GPU acceleration. *Quantum Machine Intelligence*, 8:21, 2026. doi: 10.1007/s42484-026-00365-w. arXiv:2506.23246.
- [3] Salah A. Faroughi and Farinaz Mostajeran. Neural tangent kernel analysis to probe convergence in physics-informed neural solvers: PIKANs vs. PINNs. arXiv:2506.07958, 2025.
- [4] R. Allan Freeze and John A. Cherry. *Groundwater*. Prentice-Hall, Englewood Cliffs, NJ, 1979.
- [5] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, 2018.
- [6] Said Lantigua, Gilson Giraldo, and Renato Portugal. A classical-quantum hybrid architecture for physics-informed neural networks. arXiv:2511.07216, 2025.

- [7] Jarrod R McClean, Sergio Boixo, Vadim N Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9: 4812, 2018.
- [8] M. Z. Naser. Fundamental flaws of physics-informed neural networks and explainability methods in engineering systems. *Computers & Industrial Engineering*, 212:111704, 2026. doi: 10.1016/j.cie.2025.111704.
- [9] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [10] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum machine learning models. *Physical Review A*, 103(3): 032430, 2021.
- [11] Alexandr Sedykh, Maninadh Podapaka, Asel Sagingalieva, Karan Pinto, Markus Pflitsch, and Alexey Melnikov. Hybrid quantum physics-informed neural networks for simulating computational fluid dynamics in complex shapes. *Machine Learning: Science and Technology*, 5(2): 025045, 2024. doi: 10.1088/2632-2153/ad43b2.
- [12] Matthew Tancik et al. Fourier features let networks learn high frequency functions in low dimensional domains. In *Advances in Neural Information Processing Systems*, 2020.
- [13] David Keith Todd and Larry W. Mays. *Groundwater Hydrology*. John Wiley & Sons, Hoboken, NJ, 3 edition, 2005.
- [14] Kaveh Zabihi, Hamid Montazeri, and Akke S. J. Suiker. Hybrid quantum-classical physics-informed neural networks for solving nonlinear PDEs: when and where hybridization is effective? arXiv:2606.04679, 2026.

A Figure gallery

Every figure in this appendix is drawn from committed run artifacts by `scripts/report/` (no retraining) and is also in `results/report/figures/`. Colours follow one fixed family palette, checked for colour-vision deficiency; families are also distinguished by marker and position. Circuits have one copy and runs use seeds 0–4 unless stated.

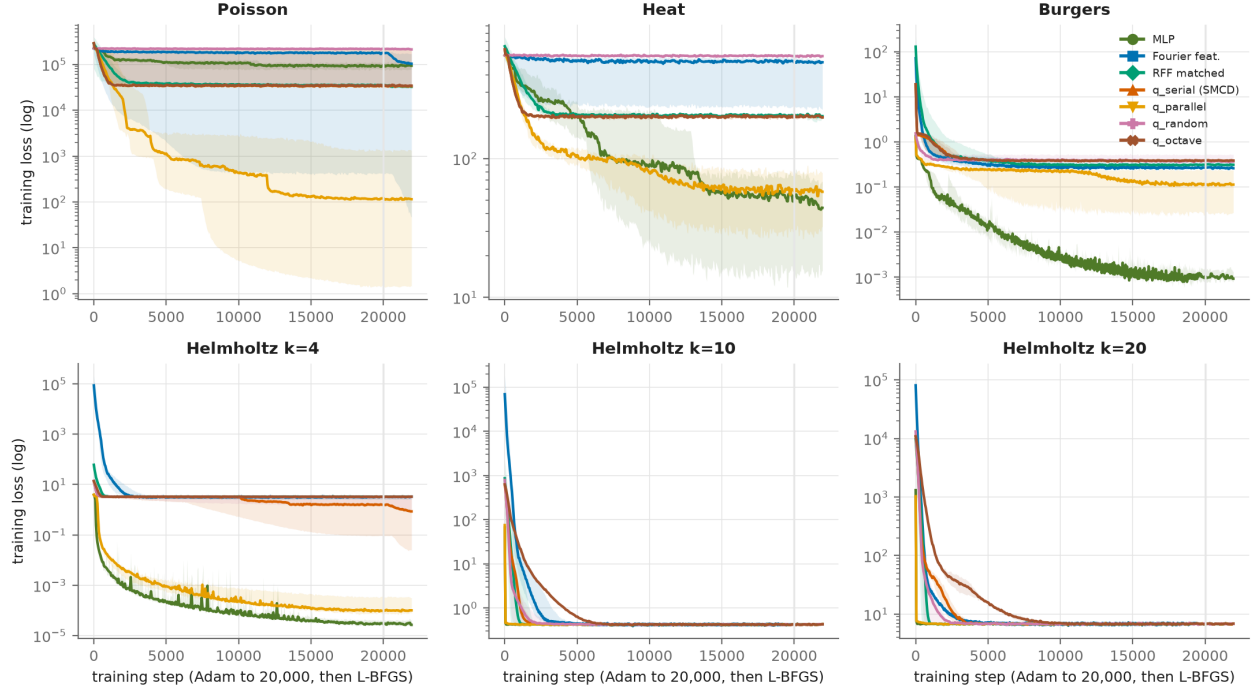


Figure 17: Training loss, median over seeds with the seed range shaded. Adam runs to step 20,000, then L-BFGS.

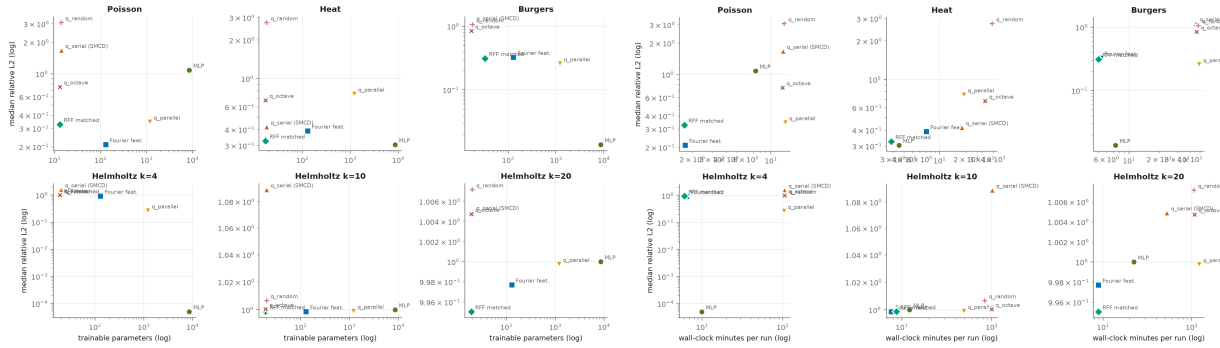


Figure 18: Median error against parameter count (left) and wall-clock time per run (right), per problem.

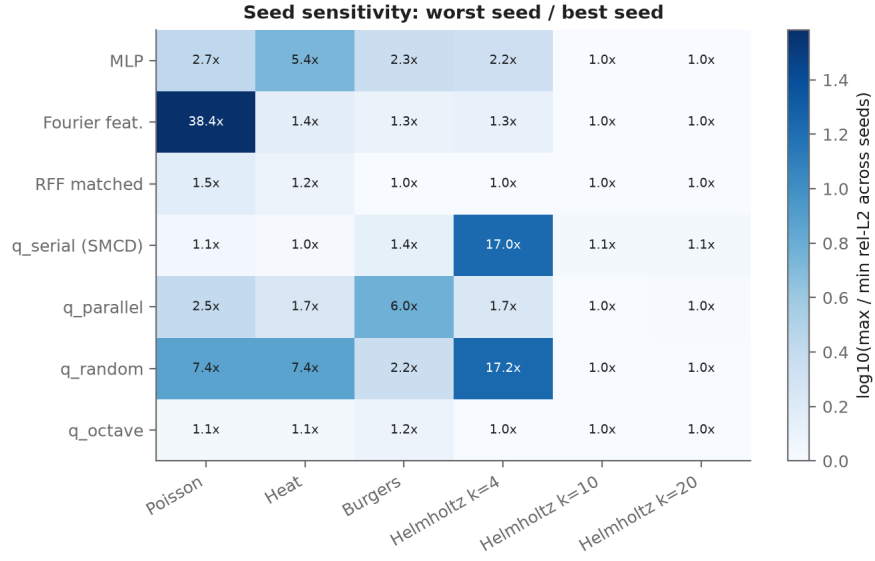


Figure 19: Seed sensitivity: worst-seed over best-seed error for each family and problem.

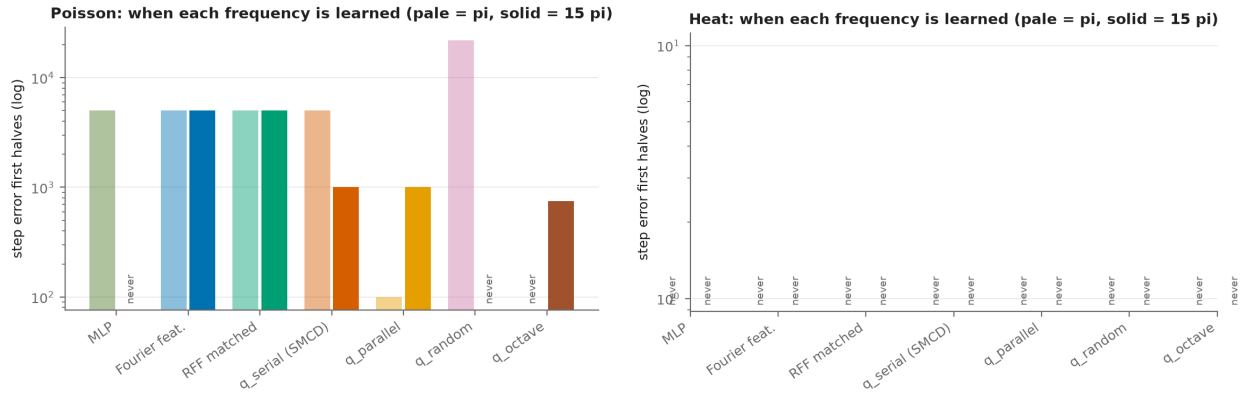


Figure 20: Step at which the error at each solution frequency first halves (median over seeds; pale: π , solid: 15π). Left: Poisson; right: heat.

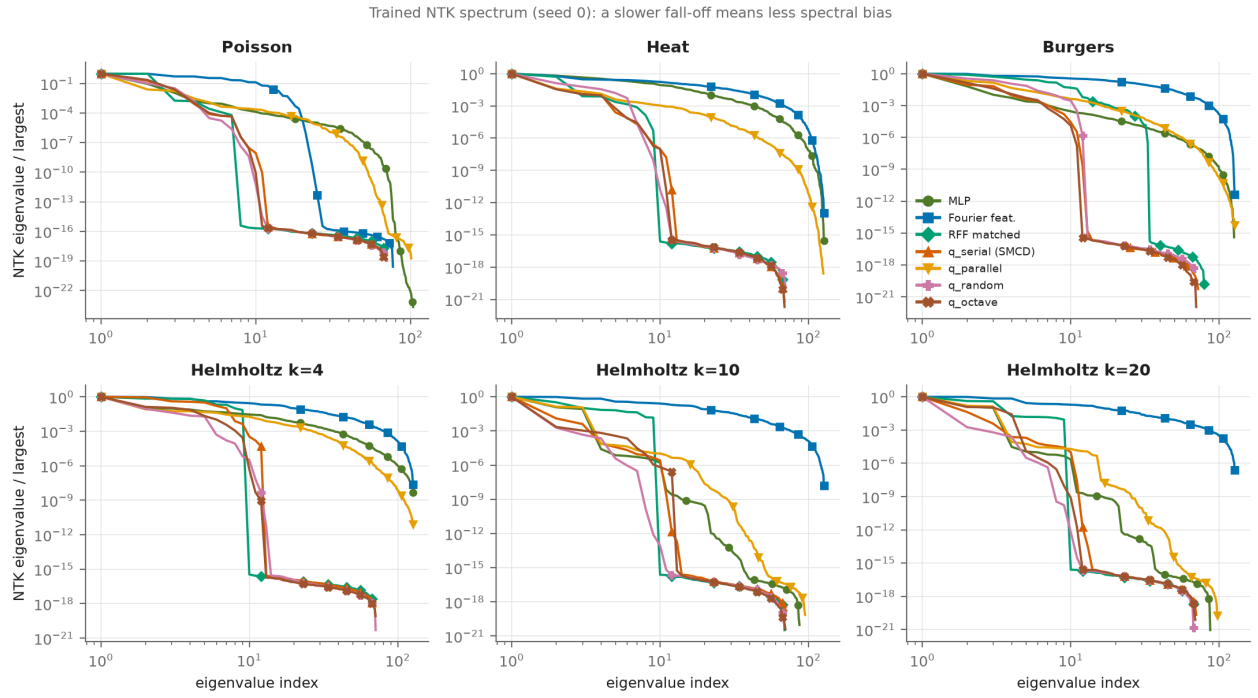


Figure 21: Trained NTK spectra (seed 0), normalised by the largest eigenvalue.

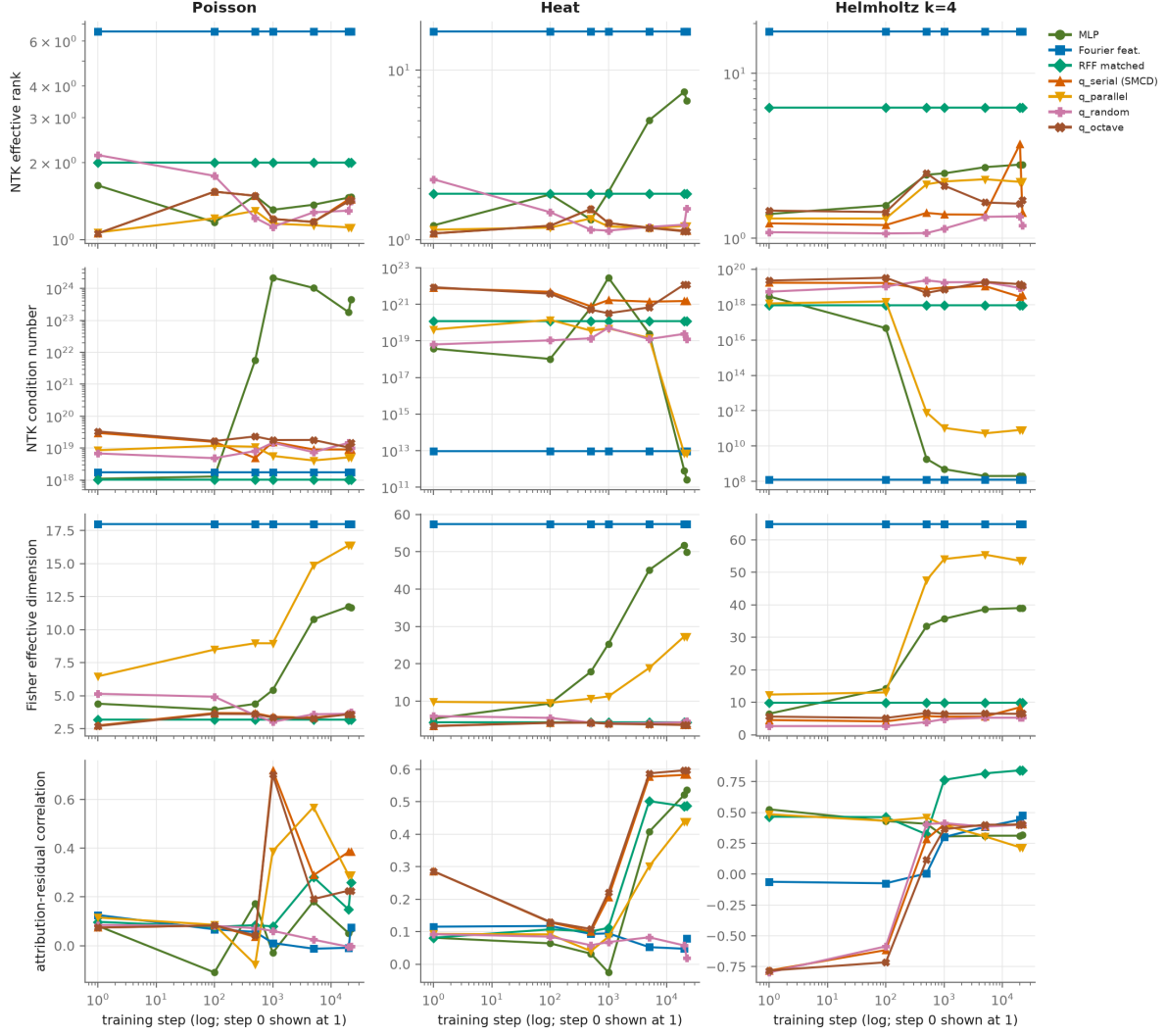


Figure 22: NTK effective rank and condition number, Fisher effective dimension and the correlation of attributions with the residual, over training (median over seeds).

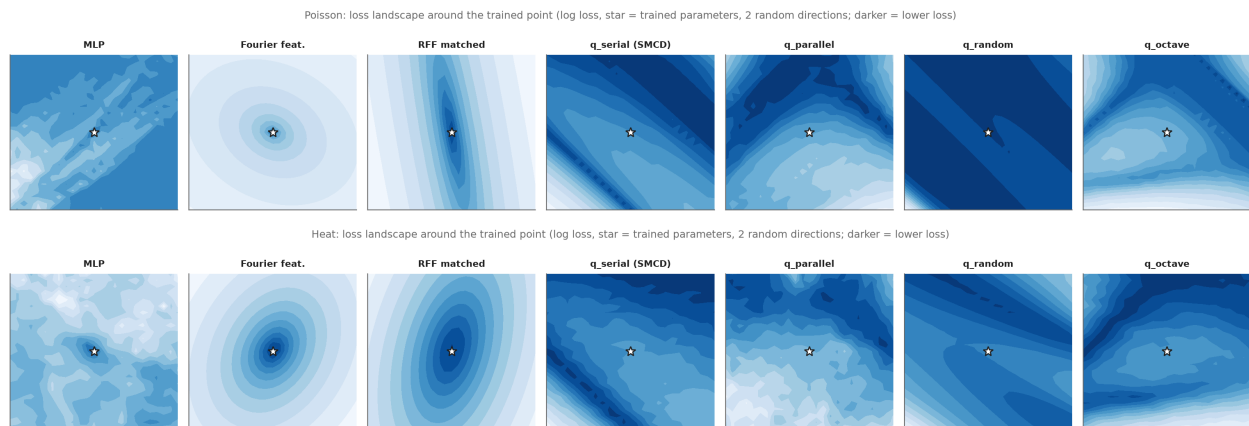


Figure 23: Loss landscape on two random directions around the trained parameters (star), Poisson (top) and heat (bottom); darker is lower loss.

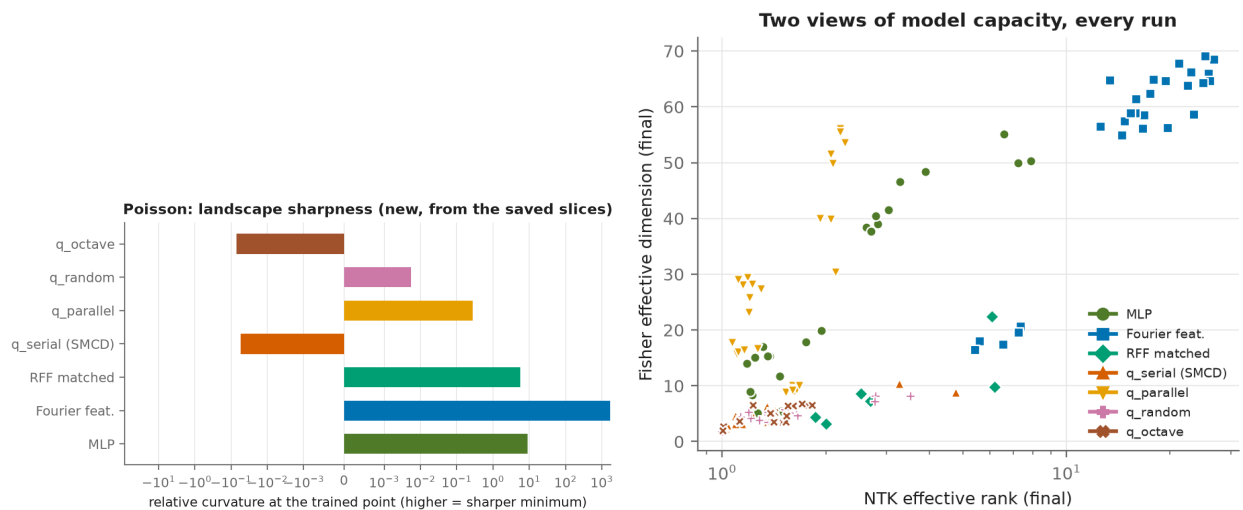


Figure 24: Left: curvature of the loss landscape at the trained point (Poisson). Right: NTK effective rank against Fisher effective dimension, every run.

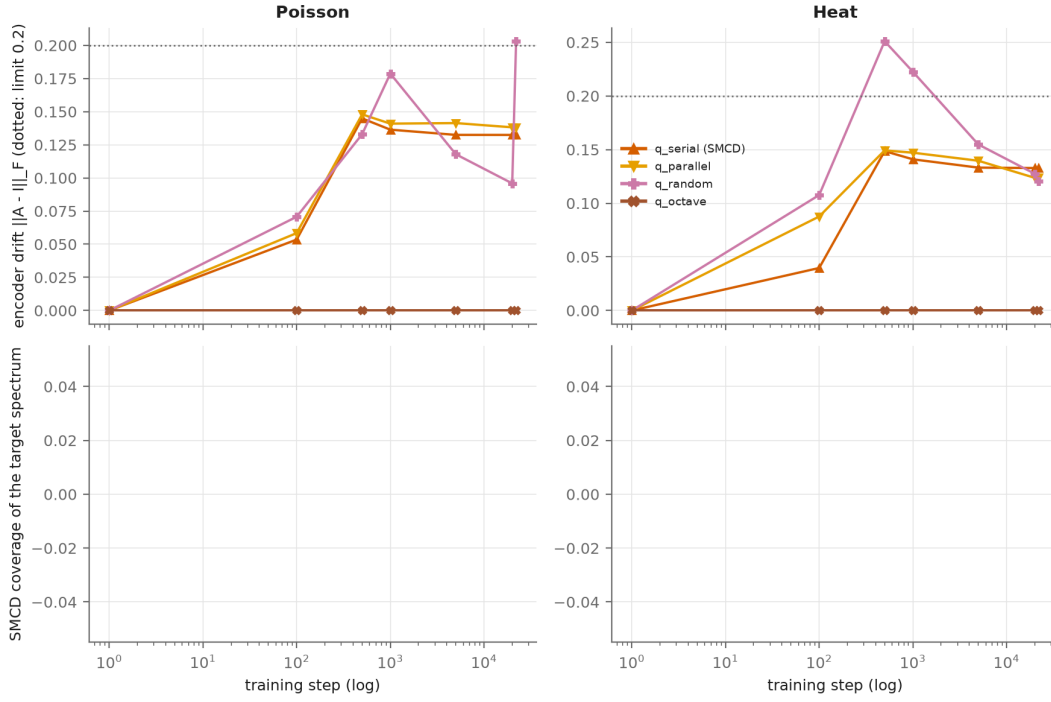


Figure 25: Quantum families: encoder drift $\|A - I\|_F$ (dotted: the 0.2 limit) and SMCD coverage of the target spectrum over training.

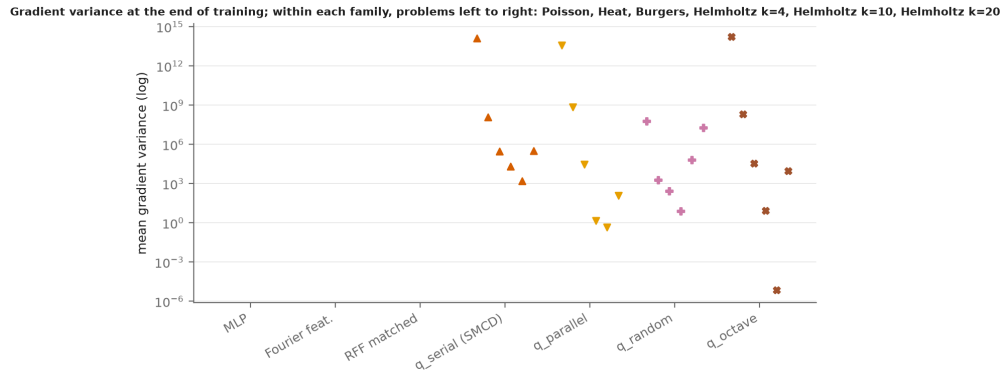


Figure 26: Mean gradient variance at the end of training, per family and problem.

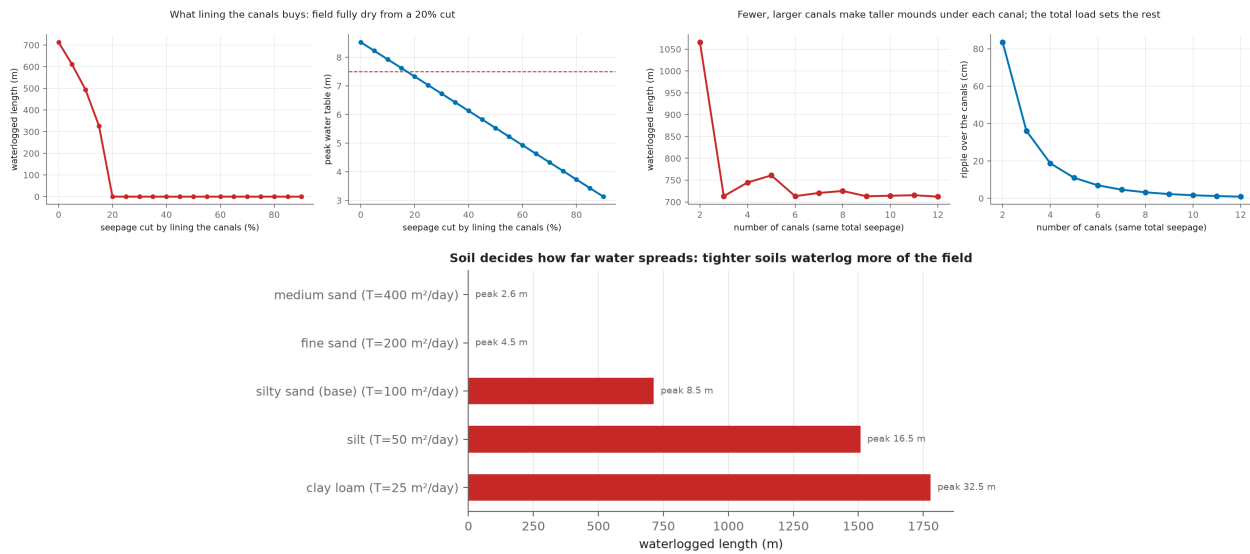


Figure 27: Groundwater scenarios from the exact solution: lining the canals, canal spacing at the same total seepage, and soil type.